



SPP PLANNING CRITERIA

Revision 5.0A

Maintained by:

TRANSMISSION WORKING GROUP
SUPPLY ADEQUACY WORKING GROUP
SYSTEM PROTECTION AND CONTROLS ADVISORY GROUP

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1. Contents

1. Contents.....	1
2. Revision History.....	6
3. Introduction.....	9
4. Planning Reserve Margin.....	10
4.1 Definitions.....	11
4.1.1 Load Responsible Entity.....	11
4.1.2 Firm Capacity.....	11
4.1.3 Peak Demand.....	12
4.1.4 Accredited Capacity.....	12
4.1.5 Summer Season.....	12
4.1.6 Winter Season.....	12
4.1.7 Electric Storage Resources.....	12
4.1.8 Demand Response Resource.....	12
4.1.9 Reliability Registered Demand Response Program (RRDRP).....	12
5. Regional Transmission Planning.....	13
5.1 Concepts.....	13
5.2 Definitions.....	13
5.3 Coordinated Planning.....	14
5.3.1 Planning Assessment Studies.....	15
5.4 SPP Compliance with NERC Reliability for Transmission Planning.....	15
5.4.1 Base Case (NERC Table 1 Planning Event “P0”).....	15
5.4.2 Bulk Electric System or Transmission System Element(s) (NERC Table 1 Planning Event P1, P2, P3, P4, P5, P6, P7).....	16
5.4.3 Extreme Events.....	16
5.4.4 System Adjustments and Mitigation Plans.....	16
5.4.5 Transmission Operating Guides.....	17
5.5 Interconnection Review Process.....	17
5.6 Electromagnetic Transient (EMT) Screening and Studies for High Impact Large Loads.....	17
6. Regional Calculation of Available Transfer Capability.....	18
6.1 Definitions.....	19
6.1.1 Base Loading, Firm and Non-Firm (FBL & NFBL).....	19

6.1.2	Capacity Benefit Margin.....	19
6.1.3	Contractual Limit.....	19
6.1.4	Critical Contingency	19
6.1.5	Designated Network Resources (DNR)	19
6.1.6	Emergency Voltage Limits	19
6.1.7	Firm Available Transfer Capability (FATC).....	19
6.1.8	First Contingency Incremental Transfer Capability.....	19
6.1.9	Flowgate.....	20
6.1.10	Line Outage Distribution Factor	20
6.1.11	Local Area Problem	20
6.1.12	Monitored Facilities	20
6.1.13	Non-firm Available Transfer Capability	20
6.1.14	Normal Voltage Limits.....	20
6.1.15	Open Access Transmission Tariff	20
6.1.16	Operating Horizon.....	21
6.1.17	Operating Procedure.....	21
6.1.18	Outage Transfer Distribution Factor	21
6.1.19	Participation Factor.....	21
6.1.20	Participation Points.....	21
6.1.21	Planning Horizon	21
6.1.22	Power Transfer Distribution Factor (PTDF).....	21
6.1.23	Power Transfer Voltage Response Factor (PTVF)	21
6.1.24	SPP Open Access Transmission Tariff (SPP OATT).....	21
6.1.25	Transfer Distribution Factor.....	21
6.1.26	Transfer Test Level	22
6.1.27	Transmission Owner	22
6.1.28	Transmission Provider.....	22
6.1.29	Transmission User	22
6.1.30	Transmission Reliability Margin (TRM)	22
6.1.31	TRM Multipliers (a & b).....	22
6.2	Concepts.....	22
6.2.1	Transfer Capability	22
6.2.2	Available Transfer Capability	23

6.2.3	Response Factors	24
6.2.4	Transfer Capability Limitations	24
6.2.5	Invalid Limits	25
6.2.6	Flowgates	26
6.2.7	Total Flowgate Capacity (TFC)	26
6.2.8	Flowgate Capacity	27
6.2.9	System Impacts	27
6.2.10	Monitored Facilities	28
6.2.11	Critical Contingencies.....	28
6.3	Reliability Margins.....	29
6.3.1	Transmission Reliability Margin (TRM).....	29
6.3.2	TRM Coordination	30
6.3.3	TRM Availability for Non-Firm Service	30
6.3.4	TRM Calculation Frequency	30
6.3.5	Capacity Benefit Margin (CBM)	31
7.	Electrical Facility Ratings	32
7.1	Accreditation and Testing Requirements	32
7.1.1	Defined Terms.....	32
7.1.2	Capability Test.....	33
7.1.3	Operational Test.....	35
7.1.4	New or Upgraded Unit Testing	36
7.1.5	Accreditation for Thermal and Hydro Units.....	36
7.1.6	Effective Load Carrying Capability (ELCC) for Solar and Wind Units.....	37
7.1.7	ELCC Accreditation for Energy Storage Units.....	40
7.1.8	Accreditation for Wind and Solar Units	41
7.1.9	Accreditation for Energy Storage Resources	42
7.1.10	Testing Requirements and Accreditation for Demand Response Resources and RRDRPS.....	42
7.1.11	Fuel Supply	45
7.2	Rating of Transmission Facilities	46
7.2.1	Power Transformer	46
7.2.2	Overhead Conductor.....	47
7.2.3	Underground Cables	51
7.2.4	Switches	53

7.2.5	Wave Wraps.....	53
7.2.6	Current Transformers	53
7.2.7	Circuit Breakers	54
7.2.8	Ratings of Series and Reactive Elements	54
7.2.9	Ratings of Energy Storage Devices.....	55
7.2.10	Facility Rating Issues	55
7.2.11	HVDC	56
8.	System Protection Requirement.....	56
8.1	Definitions.....	56
8.1.1	Effective Date.....	56
8.1.2	Equivalent Achievement of System Performance Requirement	56
8.1.3	Extended Use RAS	56
8.1.4	Temporary RAS.....	56
8.1.5	Temporary RAS Service Duration.....	57
8.1.6	RAS Sponsor	57
8.1.7	RAS Retirement.....	57
8.2	Under-Frequency Load Shedding.....	57
8.3	Remedial Action Scheme Approval.....	57
8.3.1	RAS Screening Criteria.....	57
8.3.2	RAS Approval Process	58
8.4	Under Voltage Load Shedding.....	62
9.	Uniform Rating of Generating Equipment.....	63
10.	Transmission Rating Criteria.....	65
10.1	Ambient Temperature	65
10.1.1	Purpose	65
10.1.2	Types of Ambient Temperature	65
10.1.3	Procedure for Computing Ambient Temperature	67
10.2	Underground Cables	68
10.3	Switches	70
10.4	Wave Traps	83
10.5	Current Transformers	87
10.6	Circuit Breakers.....	90
11.	Generator Reporting Forms.....	95

12.	SPP Misoperation Report.....	107
13.	Response Factor Thresholds	108
14.	Interconnection Review Process Details.....	109
14.1	Coordination	109
14.2	Prior to the Review	109
14.2.1	AC Interconnections.....	109
14.2.2	HVDC Interconnections.....	110
14.3	Technical Study Requirements.....	111
14.3.1	AC Interconnections.....	111
14.3.2	HVDC Interconnections.....	113
14.4	Dispute Resolution	113
14.5	Review and Ballot by the Transmission Working Group.....	113
14.5.1	Transmission Interconnection Review Data Checklist	113
15.	Appendix 1 - SPP RAS Thresholds.....	115
15.1	Purpose	115
15.2	RAS Thresholds	115

2. REVISION HISTORY

VERSION NUMBER	DATE	CHANGE DESCRIPTION
0.0	4/10/2015	Initial Creation
1.0	11/20/2015	Updated to reflect approval of RR 117 and RR 58_CRR014
1.1	4/26/2016	Updated to reflect approval of RR 141
1.2	4/25/2017	Updated to reflect approval of RR 186, 215, and 224
1.3	6/15/2017	Updated to reflect approval of RR 230
1.4	7/25/2017	Updated to reflect approval of RR 228
1.5	9/17/2018	Updated to reflect approval of RR 251
1.6	1/31/2019	Updated to reflect approval of RR 237
1.7	6/7/2019	Updated to reflect approval of RR 350
1.8	6/12/2019	Corrected revisions approved previously by RR 237
1.9	6/19/2019	Comprehensive document reformatting
2.0	11/5/2019	Update to reflect approval of RR 363, 364
2.1	2/18/2020	Update to reflect changes to Appendix to Sections in this SPP Planning Criteria
2.2	3/16/2020	Updated to reflect approval of RR 389
2.3	12/7/2020	Updated to reflect approval of RR 404 and 412
2.4	2/4/2021	Updated to reflect approval of RR 424
3.0	02/17/2023	Updated to reflect approval of RR 499 and 516
3.0A	02/17/2023	Updated to reflect approval of RR 452; however, language still pending FERC approval of tariff language (FERC Docket #ER23-567)
4.0	02/17/2023	Updated to reflect approval of RR 505
4.0A	02/17/2023	Version A includes language approved for RR 452; however, this tariff language is still pending FERC approval (FERC Docket #ER23-567)
4.1	05/12/2023	Planning Criteria updated to reflect FERC response related to previously approved RR462 language being inadvertently removed in last version update and modifications requested by FERC: RR 462: Added Sections 7.1.2.3, 7.1.3.4, 7.1.7

VERSION NUMBER	DATE	CHANGE DESCRIPTION
		Per FERC: 1. Removed Sections 7.1.6, 7.1.7 2. Added a new Section 7.1.6 (Note: this is the old language from previous Planning Criteria v. 2.4)
4.1A	05/12/2023	Version A includes language approved for RR452; however, this tariff language is still pending FERC approval (FERC Docket #ER23-567)
4.2	6/07/2023	Planning Criteria updated to reflect FERC approval for RR452 on 5/26/2023
4.3	1/19/2024	Planning Criteria updated to reflect MOPC and Board approval of RR572 for sections: 5.1 and 5.2
4.3A	1/19/2024	Version A includes language approved for RR554 (section 4 and 7.1) and RR568 (section 7.1); however, this tariff language is still pending FERC filing by SPP
4.4	3/29/2024	Planning Criteria updated to reflect MOPC and Board approval of RR598 for section 7.3 with an effective date of April 1, 2024
4.4A	3/29/2024	Version A includes language approved for RR554 (section 4 and 7.1) and RR568 (section 7.1); however, this tariff language is still pending FERC approval (FERC Docket #ER24-1317)
4.4A	06/30/2025	Version A includes: • language approved for RR554 (section 4 and 7.1) and RR568 (section 7.1); however, this tariff language is still pending FERC approval (FERC Docket #ER24-1317) • language approved by the Board for RR622 (section 4); however, the tariff language is still pending FERC approval (FERC Docket #ER25-89) • language approved by the Board for RR650 (sections 5.5, 7.2.11, 8, and 14); however, the tariff language is still pending FERC approval (FERC Docket #R25-2309-000)
4.5	10/24/2025	Planning Criteria updated to reflect FERC approval for RR554, RR568, RR622, and RR664 for Sections 4 & 7 with an effective date of October 1, 2025.
4.5A	10/24/2025	Version A includes: • language approved by the Board for RR650 (sections 5.5, 7.2.11, 8, and 14); however, the tariff language is still pending FERC approval (FERC Docket #R25-2309-000)

VERSION NUMBER	DATE	CHANGE DESCRIPTION
4.6	02/09/2026	Planning Criteria updated to reflect FERC approval for RR650 (sections 5.5, 7.2.11, 8, and 14) with an effective date of July 23, 2025 (FERC Docket #R25-2309-000). Updated to reflect BOD approval of RR726 (Sections 5.4 and 6.4.2.1) Planning Criteria updated to reflect FERC approval for RR696 (Sections 5.1, 5.2, and 5.6) with an effective date of 1/14/2026 (FERC Docket #: ER26-247).
4.7	04/21/2026	Planning Criteria updated to reflect FERC approval for RR607 (Section 5.2) with an effective date of 9/30/2024 (FERC Docket #ER24-2622).
4.7A	02/16/2026	Version A includes: - language approved by the Board for RR694 removing Sections 6.4 and 6.5; however, the tariff language is still pending FERC approval (FERC Docket #ER26-1605-000) - language approved by the RSC and Board for RR703 for Sections 4 and 7; however, the tariff language is still pending FERC approval (FERC Docket #ER26-2249-000) - language approved by the Board for RR720 for Sections 5.1, 5.2, 5.6, and 7; however, the tariff language is still pending FERC approval (FERC Docket #ER26-1323)
5.0	7/1/2026	Planning Criteria updated to reflect: - FERC approval for RR694 (Removing Sections 6.4 and 6.5) with an effective date of 5/4/2026 (FERC Docket #ER26-1605-000) - Board approval for RR745 (Sections 4 and 7) with an effective date of 5/5/2026.
5.0A	7/1/2026	Version A includes: - language approved by the RSC and Board for RR703 for Sections 4 and 7; however, the tariff language is still pending FERC approval (FERC Docket #ER26-2249-000) - language approved by the Board for RR720 for Sections 5.1, 5.2, 5.6, and 7; however, the tariff language is still pending FERC approval (FERC Docket #ER26-1323)

3. INTRODUCTION

The Planning Criteria developed by SPP provide background information, guidelines, business rules, and processes for the operation and administration of the SPP Planning Process.

4. PLANNING RESERVE MARGIN

The SPP East Balancing Authority Area (“SPP East BAA”) and the SPP West Balancing Authority Area (“SPP West BAA”) shall each have a separate Base Planning Reserve Margin (“Base PRM”) for purposes of compliance with Attachment AA of the SPP OATT. The SPP East BAA Base PRM shall be sixteen percent (16%) for the 2026, 2027, and 2028 Summer Season and thirty six percent (36%) for the 2026/2027, 2027/2028, and 2028/2029 Winter Season. Beginning in the Summer Season of 2029, the SPP East BAA Base PRM shall be seventeen percent (17%) for the 2029 Summer Season and thirty eight percent (38%) for the 2029/2030 Winter Season. The SPP West BAA Base PRM shall be nineteen percent (19%) for the 2027 Summer Season and forty percent (40%) for the 2027/2028 Winter Season. The Base PRM percentages for each year and each BAA are also depicted in the following table:

BAA	Summer Season	PRM %	Winter Season	PRM %
East	2026	16	2026/2027	36
East	2027	16	2027/2028	36
East	2028	16	2028/2029	36
East	2029	17	2029/2030	38
West	2027	19	2027/2028	40

The capacity used from resources when determining the Base PRM shall be the Accredited Capacity of wind, solar, and energy storage resources, excluding pumped storage hydroelectric resources, and the demonstrated net generating capability of all remaining conventional resources, Demand Response Resources, and Reliability Registered Demand Response Programs (RRDRPs) for the applicable season. Resources submitted in the East BAA will be utilized to determine the East BAA Base PRM, and resources submitted in the West BAA will be utilized to determine the West BAA Base PRM. Determination of both the Base PRM for the Summer Season and the Base PRM for the Winter Season will be supported by a probabilistic Loss of Load Expectation (“LOLE”) Study. The East BAA and West BAA will each perform separate and individual LOLE Studies to determine these Base PRM’s. The LOLE studies will analyze the ability of the Transmission Provider to reliably serve the Balancing Authority Area’s forecasted Peak Demand for the applicable season. The LOLE study for both the Summer Season and the Winter Season for each BAA will be performed in accordance with Attachment AA of the SPP OATT. Prior to June 1, 2026, if a Load Responsible Entity’s (“LRE”) Firm Capacity for the Summer Season is comprised of at least seventy-five percent (75%) hydro-based generation, then such Base PRM shall be nine point eight nine percent (9.89%) for the Summer Season. Prior to June 1, 2026, if a LRE’s Firm Capacity for the Winter Season is comprised of at least seventy-five percent (75%) hydro-based generation, then such Base PRM shall be nine point eight nine percent (9.89%) for the Winter Season.

Commented [SP1]: RR703 Demand Response

- Approved by the RSC on:
- Approved by the Board on:
- Posted with FERC: 4/17/2026
 - FERC Docket #: ER26-2249-000
- Requested FERC Effective Date: 1/1/2027

Effective June 1, 2026, the Base PRM for the Summer Season and the Base PRM for the Winter Season will be used as a basis for calculating an Accredited Capacity PRM for the applicable season by applying the Accredited Capacity for the applicable season in accordance with Attachment AA of the Tariff. The difference of ~~the~~ conventional resources', Demand Response Resources', or RRDRPs' demonstrated net generating capability and the Accredited Capacity for the applicable season will be used to calculate the Accredited Capacity PRM ~~for the applicable season~~. Both the Base PRM and Accredited Capacity PRM include the Accredited Capacity for run-of-the-river hydroelectric resources, wind resources, solar resources, and Electric Storage Resources, excluding pump storage hydroelectric resources for the applicable season and BAA. The example below is given for illustration purposes. Effective June 1, 2026, for the East BAA, the calculated Accredited Capacity PRM for the Summer Season and the Accredited Capacity PRM for the Winter Season shall be applied to all Load Responsible Entities, regardless of resource mix. Effective June 1, 2027, for the West BAA, the calculated Accredited Capacity PRM for the Summer Season and the Accredited Capacity PRM for the Winter Season shall be applied to all Load Responsible Entities, regardless of resource mix.

Variable	Value	Formula
Summation of LRE <u>forecasted</u> Peak Demand	50,000	[A]
Base PRM	15%	[B]
Minimum Required Capacity (Base PRM)	57,500	[C] = [A] + [A]*[B]
Summation of <u>Demonstrated Net Generating Capability from</u> Conventional Resources, <u>Demonstrated Net Generating Capability Demand Response Resources, and RRDRPs</u>	60,000	[D]
Summation of <u>Accredited Capacity from</u> Conventional Resources, <u>Demand Response Resources, and RRDRPs Accredited Capacity</u>	57,000	[E]
Difference in <u>capacity from</u> Conventional Resources, <u>Demand Response Resources, and RRDRPs Capacity</u>	3,000	[F] = [D] - [E]
Minimum Required Accredited Capacity	54,500	[G] = [C] - [F]
Accredited Capacity PRM	9%	[H] = ([G]-[A]) / [A]

Table 4.1: Is used for reference purposes only and does not always reflect the latest PRM.

4.1 DEFINITIONS

4.1.1 **LOAD RESPONSIBLE ENTITY**

As defined in Attachment AA of the SPP OATT.

4.1.2 **FIRM CAPACITY**

As defined in Attachment AA of the SPP OATT.

4.1.3 **PEAK DEMAND**

As defined in Attachment AA of the SPP OATT.

4.1.4 **ACCREDITED CAPACITY**

As defined in Attachment AA of the SPP OATT.

4.1.5 **SUMMER SEASON**

As defined in Attachment AA of the SPP OATT.

4.1.6 **WINTER SEASON**

As defined in Attachment AA of the SPP OATT.

4.1.7 **ELECTRIC STORAGE RESOURCES**

As defined in Attachment AA of the SPP OATT.

4.1.8 **DEMAND RESPONSE RESOURCE**

As defined in Attachment AA of the SPP OATT.

4.1.9 **RELIABILITY REGISTERED DEMAND RESPONSE PROGRAM (RRDRP)**

As defined in Attachment AA of the SPP OATT.

Commented [SP2]: RR703 Demand Response

- Approved by the RSC on:
- Approved by the Board on:
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- Requested FERC Effective Date: 1/1/2027

5. REGIONAL TRANSMISSION PLANNING

5.1 CONCEPTS

For the purposes of Section 5 of the SPP Criteria the transmission system shall be defined as facilities under the functional control of the SPP Open Access Transmission Tariff (OATT) or the Bulk Electric System (BES). The transmission system shall be capable of performing reliably under a wide variety of expected system conditions while continuing to operate within equipment and electric system thermal, voltage, and stability limits. The transmission system, at a minimum, shall be planned to withstand all single element contingencies and maintenance outages over the load conditions of all applicable seasonal models as required for each planning process. Extreme event contingencies which measure the robustness of the electric systems should be evaluated for risks and consequences. The NERC Reliability Standards define specific requirements where adherence provides a measurable degree of reliability for the BES. SPP provides additional coordinated regional transmission planning requirements to promote reliability through this Criterion and related "Transmission Planning Process" (Attachment O) in the OATT. As recognized in Attachment O, SPP is responsible for assessing requests for interconnection of generation facilities to the transmission system (Attachment V and Attachment BB of the OATT) and for assessing upgrades (ITP), retirement (Attachment AB of the OATT), replacement (Attachment V of the OATT), or establishment of delivery points (Attachment ~~AQ-04~~, Attachment AX, or Attachment BC) of the OATT).

5.2 DEFINITIONS

All capitalized terms shall have their meaning as defined in the SPP OATT or NERC's Glossary of Terms (and as used in the NERC Reliability Standards), unless defined below or noted within this document.

The definition of Transmission Material Modification is used for purposes of evaluating changes to existing non-radial facilities placed under the functional control of the SPP Regional Transmission Organization (non-radial Transmission System facility) and any additional BES interconnections of a transmission facility. The definition will be used for any tariff process that references Transmission Material Modification. The definition of Qualified Change is used for purposes of compliance with the requirements of NERC's mandatory reliability standard FAC-002-04¹, or any other NERC mandatory reliability standard that refers to a "qualified change." If one or more Transmission Material Modification or Qualified Change criterion are met, SPP shall analyze these changes to meet the requirements of SPP OATT processes as the Transmission Provider and of any NERC mandatory reliability standard as the Planning Coordinator. Any change outside of this definition may be submitted to the Planning Coordinator for evaluation.

Nominal Voltage – The root-mean-square, phase-to-phase voltage by which the system is designated and to which certain operating characteristics of the system are related. Examples of nominal voltages are 500 kV, 345 kV, 230 kV, 161 kV, 138 kV, 115 kV and 69 kV.

¹ FAC-002-04 shall mean the current effective version of the Reliability Standard, and shall not be considered a reference a standard subject to future enforcement or revisions subject to current NERC project development.

Commented [SP3]: RR720 Conditional High Impact Large Loads

- Approved by the Board on: 2/3/2026
- Posted with FERC: 2/10/2026
- FERC Docket #: ER26-1323
- Requested FERC Effective Date: 7/1/2026

"SPP submitted revisions to the Tariff to add proposed Conditional High Impact Large Load Service ("CHILLS") that will allow High Impact Large Loads ("HILLS") to be served with non-firm conditional transmission service. On March 20, 2026, SPP filed an answer in response to comments and the limited protest filed in this proceeding. On March 30, 2026, FERC issued a letter stating that SPP's filing is deficient, and additional information is necessary to process the filing."

Qualified Change – A permanent change (that is typically greater than 12 months in duration) to an existing BES interconnections of transmission, generation, or electricity end-user Facility, pursuant to the following criteria:

- 1) *Transmission Qualified Change* – A “Transmission Material Modification,” as defined in this Section 5.2 of the SPP Planning Criteria, and further described in Sections 5.4 and 5.5 of this SPP Planning Criteria.
- 2) *Generation Qualified Change* – A change to existing generation or a Generator Interconnection Request or its Interconnection Facilities, pursuant to Attachment V and Attachment BB of the SPP OATT. Any changes studied and deemed a “Material Modification,” as defined in Section 1 of Attachment V of the SPP OATT, are subject to the requirements of Attachment V or Attachment BB, respectively.
- 3) *Electricity End-User Qualified Change* – A change in delivery point facilities, to upgrade, retire, replace or establish a delivery point, including metering or other facilities at such location, or for unexpected step load changes at an existing delivery point at the discretion of the host Transmission Owners in accordance with Attachment ~~AQ-01~~, Attachment AX, or Attachment BC of the SPP OATT.

System Peak Responsibility - A Transmission Customer’s projected load as provided in accordance with Section 17.8 or 29.2 of the SPP OATT.

Transmission Material Modification- A permanent change (that is typically greater than 12 months) to BES transmission or SPP OATT facilities. These permanent changes include:

- 1) Reduction to a non-radial Transmission System facility or BES transmission facility’s normal rating or emergency rating greater than 20% (derate);
- 2) Proportional changes to the magnitude of the non-radial Transmission System facility or BES facility’s impedance that is greater than +/- 30% from its original positive sequence impedance value;
- 3) Changes in operating voltage of a non-radial Transmission System or BES transmission facility;
- 4) Changes in system configuration including the connection of new non-radial Transmission facilities or BES facilities or disconnection of existing non-radial Transmission facilities or BES facilities;
- 5) Changes in system protection that would reduce fault-interrupting capability or fault-clearing expediency for events that are included in the SPP Annual data request.

5.3 COORDINATED PLANNING

SPP members operate in a highly interconnected transmission system and shall coordinate transmission planning. This coordination shall include efforts between interconnected SPP members and non-members. SPP shall be the primary responsible party for coordinated transmission planning.

The planning and development of the transmission system shall be coordinated with neighboring systems and regions to preserve the reliability benefits of interconnected operations. The transmission systems should be planned to avoid unacceptable system

Commented [SP4]: RR720 Conditional High Impact Large Loads

- Approved by the Board on: 2/3/2026
- Posted with FERC: 2/10/2026
- FERC Docket #: ER26-1323
- Requested FERC Effective Date: 7/1/2026

“SPP submitted revisions to the Tariff to add proposed Conditional High Impact Large Load Service (“CHILLS”) that will allow High Impact Large Loads (“HILLS”) to be served with non-firm conditional transmission service. On March 20, 2026, SPP filed an answer in response to comments and the limited protest filed in this proceeding. On March 30, 2026, FERC issued a letter stating that SPP’s filing is deficient, and additional information is necessary to process the filing.”

performance as described in Section 5.4 based on any one transmission circuit, structure, right-of-way, or substation.

SPP staff and applicable stakeholder groups (at a minimum, the Model Development Working Group and Transmission Working Group) shall coordinate to verify power flow models, short-circuit models, and stability models, which shall be used by SPP staff to comply with NERC Reliability Standards and for studies as required by the OATT. Extreme contingency evaluations shall be conducted to measure the robustness of the transmission system and to maintain a state of preparedness to deal effectively with such events. Although it is not practical to construct a system to withstand all possible extreme contingencies, it is desirable to understand the risks and consequences of such events and to attempt to limit the significant economic and social impacts that may result.

5.3.1 PLANNING ASSESSMENT STUDIES

Individual Transmission Owners under the OATT shall perform transmission planning studies required by the OATT and shall cooperate within the SPP Transmission Expansion Plan and other SPP coordinated studies. These planning studies are for the purposes of identifying any planning criteria violations that may exist and developing plans to mitigate such violations. Members shall contact the SPP and the TWG whenever new facilities are in the planning stage so that optimal integration of any new facilities and potentially impacted parties can be identified. Studies affecting more than one system owner or user will be conducted on a joint coordinated system basis. Reliability studies shall examine system intact and post-contingency conditions to identify unacceptable system performance described in Section 5.4. Updates to the transmission assessments will be performed, as appropriate, to reflect anticipated significant changes in system conditions.

5.4 SPP COMPLIANCE WITH NERC RELIABILITY FOR TRANSMISSION PLANNING

For assessments of the transmission system performed in accordance with the OATT, some planning events described in Section 5.4 may not be evaluated. Additionally, assessments performed in accordance with the OATT may evaluate additional contingencies not included in the BES, but included under the functional control of SPP, including 69 kV facilities. More information regarding the specific contingencies studied can be found in the scoping documents of the applicable assessment, including but not limited to the OATT, SPP OATT Business Practices, or ITP Manual. SPP requires that assessments verify the transmission system within the planning models conform to the following performance criteria as applicable. This criteria will apply to the SPP West Region and SPP East Region unless stated differently in the sections below:

5.4.1 BASE CASE (NERC TABLE 1 PLANNING EVENT “P0”)

- 1) Facility² loadings within the Normal Rating per SPP Planning Criteria 7.
- 2) System steady state voltages within plus or minus five percent (+/- 5%) of Nominal Voltage, except for those entities whose Normal Rating is within their applicable facility

² The capitalization of this term is included for grammatical purposes only.

ratings.

- 3) System Stability – All BES generators shall remain in synchronism. Dynamic, transient, and steady state stability of the transmission system shall be maintained. A no-fault 20 second simulation, from an industry recognized software tool, shall have a flat response with respect to machine speed, angle, power, and VARs for all machines in the transmission system.

5.4.2 **BULK ELECTRIC SYSTEM OR TRANSMISSION SYSTEM ELEMENT(S)** **(NERC TABLE 1 PLANNING EVENT P1, P2, P3, P4, P5, P6, P7)**

- 1) Facility³ loadings within the Emergency Rating per SPP Planning Criteria 7.
- 2) System steady state voltages within plus five or minus ten percent (+5/-10%) of Nominal Voltage on the BES or on load serving buses under the functional control of the SPP OATT, as applicable in each study process, except for those entities whose Emergency Rating is within their applicable facility ratings.
- 3) System Stability - All BES generators shall remain in synchronism. Machine Rotor Angles shall exhibit well damped angular oscillations and proper voltage response following a disturbance on the transmission system in accordance with the Southwest Power Pool Disturbance Performance Requirements.

5.4.3 **EXTREME EVENTS**

An extreme event shall have the meaning consistent with NERC Reliability Standard TPL-001 Table 1 – Steady State & Stability Performance Extreme Events. SPP shall run contingency studies as provided by the Transmission Owners under the following conditions:

- 1) Initiating event(s) shall result in multiple elements out of service.
- 2) SPP shall document the measures and procedures to mitigate or eliminate the extent and effects of those events and may at their discretion recommend such measures and procedures where extreme contingency events could lead to uncontrolled cascading outages or system instability.

5.4.4 **SYSTEM ADJUSTMENTS AND MITIGATION PLANS**

When simulations indicate unacceptable system performance as described in Section 4.4, the applicable Transmission Owner must provide a written summary of their system adjustments or mitigation plans, including a schedule for implementation, to achieve the required system performance throughout the planning horizon. Mitigation plan summaries should discuss expected required in-service dates of facilities, should consider lead-times necessary to implement plans, and will be reviewed for continuing need in subsequent annual assessments.

³ The capitalization of this term is included for grammatical purposes only.

5.4.5 **TRANSMISSION OPERATING GUIDES**

A transmission operating guide qualifies as a valid mitigation measure when the transmission operating guide is effective as written.

5.5 **INTERCONNECTION REVIEW PROCESS**

Southwest Power Pool Planning Criteria 5.3.1 and the SPP Open Access Transmission Tariff both require members to contact SPP and the Transmission Working Group whenever new transmission facilities that impact the interconnected operation are in the conceptual planning stage so that the optimal integration of any new facilities can be identified. Under this criterion an interconnection involves two or more SPP members or an SPP member and a non-member. A project that (1) creates a non-radial, non-generation alternating current (AC) interconnection at 69 kV or above or that removes an interconnection at 230 kV or above, or (2) is a new high-voltage direct current (HVDC) interconnection, shall be reviewed for impacts in accordance with Section 14 of this Criteria.

5.6 **ELECTROMAGNETIC TRANSIENT (EMT) SCREENING AND STUDIES FOR HIGH IMPACT LARGE LOADS**

All High Impact Large Loads (HILL) as defined in the SPP Tariff will require a short-circuit ratio and critical clearing time (SCRCCT) analysis to determine if detailed EMT analysis is needed.

If detailed EMT analysis is needed SPP may take into account the following:

- HILL frequency response and damping.
- Control instability and sub synchronous interactions.
- Interaction with grid-following or grid-forming inverter based resources.
- Harmonic interactions.
- Transients such as Fault-Induced Delayed Voltage Recovery (FIDVR).

6. REGIONAL CALCULATION OF AVAILABLE TRANSFER CAPABILITY

SPP takes a regional approach in the determination of Available Transfer Capability (ATC). The regional approach calls for SPP to evaluate the inter-area transfer capability of its Transmission Owners. This approach provides a high level of coordination between ATC reported by SPP and Transmission Owners on SPP Open Access Same-time Information Network (OASIS) nodes. Likewise, when Transmission Owners calculate ATC, they are responsible to coordinate the ATC between their areas. If there is a dispute concerning the ATC, the SPP Transmission Working Group (TWG) will act as the technical body to determine the ATC to be reported. This Planning Criteria provides Transmission Owners and the SPP Transmission Provider flexibility to revise the ATC as needed for changes in operating conditions, while providing for unique modeling parameters of the areas. The SPP Transmission Provider calculations do not preclude any studies made by Transmission Owners in accordance with their individual tariffs, which may contain specific methodologies for evaluating transmission service requests.

Transfer capabilities are calculated for two different commercial business applications; a) for use as default values for Transmission Owners to post on their OASIS node for business under their transmission tariffs and b) for use by SPP in administering the SPP Open Access Transmission Tariff (SPP OATT).

The SPP utilizes a “constrained element” approach in determining ATC. This approach is referred to as a Flowgate ATC methodology. Constrained facilities, termed “Flowgates”, used in this approach are identified primarily from a non-simultaneous transfer study using standard incremental transfer capability techniques that recognize thermal, voltage and contractual limitations. Stability limitations are studied as needed. Flowgates serve as proxies for the transmission network and are used to study system response to transfers and contingencies. Using Flowgates with pre-determined ratings, this process is able to evaluate the ATC of specific paths on a constrained element basis (Flowgate basis) while considering the simultaneous impact of existing transactions.

The calculation of ATC is a very complex and dynamic procedure. SPP realizes that there are many technical and policy issues concerning the calculation of ATC that will evolve with industry changes. Therefore, the SPP Operating Reliability Working Group and the SPP Transmission Working Group will have the joint authority to modify the implementation of this Section of the Criteria based on experience and improvements in technology and data coordination. Any changes made by these groups will be subject to formal approval as outlined in the SPP Bylaws at the first practical opportunity, with the exception of response factor thresholds for short-term transmission service which may be approved for immediate implementation by the ORWG subject to subsequent review by the MOPC at the first practical opportunity. The response factor thresholds for short-term and long-term service are included in Section 13.

6.1 DEFINITIONS

6.1.1 **BASE LOADING, FIRM AND NON-FIRM (FBL & NFBL)**

The determined loading on a Flowgate resulting from the net effect of modeled existing transmission service commitments for the purpose of serving firm network load and impacts from existing OATT OASIS commitments

6.1.2 **CAPACITY BENEFIT MARGIN**

The amount of Flowgate capacity reserved by load serving entities to ensure access to generation from interconnected systems to meet generation reliability requirements

6.1.3 **CONTRACTUAL LIMIT**

Contractual arrangements between Transmission Providers that define transfer capability between the two

6.1.4 **CRITICAL CONTINGENCY**

Any generation or transmission facility that, when outaged, is deemed to have an adverse impact on the reliability of the transmission network

6.1.5 **DESIGNATED NETWORK RESOURCES (DNR)**

Any designated generation resource that can be called upon at any time for the purpose of serving network load on a non-interruptible basis. The designated generation resource must be owned, purchased or leased by the owner of the network load

6.1.6 **EMERGENCY VOLTAGE LIMITS**

The operating voltage range on the interconnected system that is acceptable for the time sufficient for system adjustments to be made following a Critical Contingency

6.1.7 **FIRM AVAILABLE TRANSFER CAPABILITY (FATC)**

The determined transfer capability available for firm Transmission Service as defined by the FERC pro forma OATT or any direction of interest on a transmission network between generation groups and/or system load for which commercial service may be desired.

6.1.8 **FIRST CONTINGENCY INCREMENTAL TRANSFER CAPABILITY**

NERC Transmission Transfer Capability, reference document (May 1995) defines FCITC as: "The amount of power, incremental and above normal base transfers, that can be transferred over the interconnected transmission systems in a reliable manner based on all of the following conditions:

- 1) For the existing or planned system configuration, and with normal (pre-contingency) operating procedures in effect, all facility loadings are within normal ratings and all voltages are within normal limits,

- 2) The electric systems are capable of absorbing the dynamic power swings, and remaining stable, following a disturbance that results in the loss of any single electric system element, such as a transmission circuit, transformer or generating unit, and,
- 3) After the dynamic power swings subside following a disturbance that results in the loss of any single electric system element as described in 2 above, and after the operation of any automatic operating systems, but before any post-contingency operator-initiated system adjustments are implemented, all transmission facilities loadings are within emergency ratings and all voltages are within emergency limits."

6.1.9 **FLOWGATE**

A selected transmission element or group of elements acting as proxy for the transmission network representing potential thermal, voltage, stability and contractual system constraints to power transfer. The process of determining the reliability issues for which a Flowgate is representative of and by which a Flowgate is established is outlined in SPP Planning Criteria Section 6.4.

6.1.10 **LINE OUTAGE DISTRIBUTION FACTOR**

The percent of the power flowing across the contingency facility that transfers over the monitored facility when the contingency facility is switched out of service.

6.1.11 **LOCAL AREA PROBLEM**

A Transmission Owner may declare a facility under its control a Local Area Problem if it is overloaded in either the base case or contingency case prior to the transfer. If a member declares a facility a Local Area Problem, the member may neither deny transmission service nor request NERC Transmission Loading Relief for that defined condition.

6.1.12 **MONITORED FACILITIES**

Any transmission facility that is checked for predefined transmission limitations.

6.1.13 **NON-FIRM AVAILABLE TRANSFER CAPABILITY**

The determined transfer capability available for sale for non-firm Transmission Service as defined by the FERC pro forma OATT for any direction of interest on a transmission network between generation groups and/or system load for which commercial service may be desired.

6.1.14 **NORMAL VOLTAGE LIMITS**

The operating voltage range on the interconnected system that is acceptable on a sustained basis.

6.1.15 **OPEN ACCESS TRANSMISSION TARIFF**

FERC approved Pro-Forma Open Access Transmission Tariff.

6.1.16 **OPERATING HORIZON**

Time frame for which Hourly transmission service is offered. The rolling time frame is twelve to 36 hours with a 12 noon threshold. It includes the current day, and after 12 noon, the remainder of the current day and all hours of the following day.

6.1.17 **OPERATING PROCEDURE**

Any policy, practice or system adjustment that may be automatically implemented, or manually implemented by the system operator within a specified time frame, to maintain the operational integrity of the interconnected electric systems. If an Operating Procedure is submitted to the SPP in writing and states that it is an unconditional action to implement the procedure without regard to economic impacts or existing transfers, then the Operating Procedure will be used to allow transfers to a higher level.

6.1.18 **OUTAGE TRANSFER DISTRIBUTION FACTOR**

The percentage of a power transfer that flows through the monitored facility for a particular transfer when the contingency facility is switched out of service.

6.1.19 **PARTICIPATION FACTOR**

The percentage of the total power adjustment that a participation point will contribute when simulating a transfer.

6.1.20 **PARTICIPATION POINTS**

Specified generators that will have their power output adjusted to simulate a transfer.

6.1.21 **PLANNING HORIZON**

Time frame beyond which Hourly transmission service is not offered.

6.1.22 **POWER TRANSFER DISTRIBUTION FACTOR (PTDF)**

The percentage of power transfer flowing through a facility or a set of facilities for a particular transfer when there are no contingencies.

6.1.23 **POWER TRANSFER VOLTAGE RESPONSE FACTOR (PTVF)**

The per unit amount that a facility's voltage changes due to a particular transfer level.

6.1.24 **SPP OPEN ACCESS TRANSMISSION TARIFF (SPP OATT)**

The Southwest Power Pool Regional FERC approved Open Access Transmission Tariff

6.1.25 **TRANSFER DISTRIBUTION FACTOR**

A general term, which may refer to either PTDF or OTDF - The TDF represents the relationship between the participation adjustment of two areas and the Flowgates within the system.

6.1.26 **TRANSFER TEST LEVEL**

The amount of power that will be transferred to determine facility TDFs for use in DC linear analysis.

6.1.27 **TRANSMISSION OWNER**

An Entity that owns transmission facilities which are operated under a FERC approved OATT

6.1.28 **TRANSMISSION PROVIDER**

An entity responsible for administering a transmission tariff. In the case of the SPP OATT, SPP is the Transmission Provider. An SPP member may be its own Transmission Provider if the member continues to sell transmission service under the terms of its own tariff.

6.1.29 **TRANSMISSION USER**

Any entities that are parties to transactions under appropriate tariffs.

6.1.30 **TRANSMISSION RELIABILITY MARGIN (TRM)**

The amount of Flowgate capacity necessary to ensure that the interconnected transmission network is secure under a reasonable range of uncertainties in system conditions.

6.1.31 **TRM MULTIPLIERS (A & B)**

- 1) "a"-multiplier; a factor between 0 and 1 indicating the amount of TRM not available for non-firm use during the Planning Horizon
- 2) "b"-multiplier; a factor between 0 and 1 indicating the amount of TRM not available for non-firm use during the Operating Horizon

6.2 **CONCEPTS**

6.2.1 **TRANSFER CAPABILITY**

Transfer capability is the measure of the ability of the interconnected electric systems to reliably move or transfer power from one area to another over all transmission circuits (or paths) between those areas under specified system conditions. The units of transfer capability are in terms of electric power, generally expressed in megawatts (MW). Transfer capability is also directional in nature. That is, the transfer capability from area A to area B is not generally equal to the transfer capability from area B to area A.

Some major points concerning transfer capability analysis are briefly outlined below:

- 1) System Conditions - Base system conditions are identified and modeled for the period being analyzed, including projected customer demand, generation dispatch, system configuration and base reserved and scheduled transfers.
- 2) Critical Contingencies - During transfer capability studies, both generation and transmission system contingencies are evaluated to determine which facility outages are most restrictive to the transfer being analyzed.

- 3) System Limits - The transfer capability of the transmission network can be limited by thermal, voltage, stability or contractual considerations.

Thermal and voltage transfer limits can be determined by calculating the First Contingency Incremental Transfer Capability. Stability studies may be performed by the Transmission Owners at their discretion. Any known stability limits, which are determined on a simultaneous basis, and all contractual limits will be supplied by each Transmission Owner in writing to the Transmission Provider and the TWG.

6.2.2 **AVAILABLE TRANSFER CAPABILITY**

NERC *Available Transfer Capability Definitions and Determinations*, reference document (June 1996) states: "Available Transfer Capability (ATC) is a measure of the transfer capability remaining in the physical transmission network for further commercial activity over and above already committed uses."

SPP determines ATC as a function of the most limiting Flowgate of the path of interest. How limiting a Flowgate is to a path is based on two aspects: (1) The determined firm or non-firm Available Flowgate Capacity (FAFC or NFAFC) for that Flowgate, and (2) the TDF for which that Flowgate responds to power movement on the path under evaluation.

The common relationship between the identified limiting Flowgate and the path is the Transfer Distribution Factor (TDF). This is mathematically expressed as follows:

- 1) Firm ATC = the firm Available Flowgate Capacity divided by the Transmission Distribution Factor ($FATC = FAFC/TDF$) of the associated path.

Likewise,

- 2) Non-Firm ATC = the non-firm Available Flowgate Capacity divided by the Transmission Distribution Factor ($NFATC = NFAFC/TDF$) of the associated path.

Path ATC is determined by identifying the most limiting Flowgates to the path in question. Each Flowgate represents a potential limiting element to any path within a system. Therefore, each Flowgate with known Transfer Distribution Factor (TDF) can be translated into path ATC.

However, the Flowgate that produces the most limiting path ATC is the key Flowgate for that path.

The calculation of path ATC using this method is based on the ratio of the TDF into the remaining capacity of a Flowgate, (non-firm Available Flowgate Capacity or firm Available Flowgate Capacity). Once a group of potential limiting elements has been selected, then all values pertaining to ATC can be translated based on the TDF.

6.2.3 **RESPONSE FACTORS**

Response Factors are numerical relationships between key adjustments in the transmission system and specific transmission components being monitored. They provide a linear means of extrapolation to an anticipated end for which decisions can be made. The thresholds for several of the following response factors are listed in Section 13.

- 1) Transfer Distribution Factor - The Transfer Distribution Factor (TDF) is a general term referring to either PTDF or OTDF. The relationship between adjustments in participation points associated with a specific path and the identified Flowgate in the system is the TDF. Depending on the Flowgate type, the TDF may specifically represent the response in the system to certain types of pre-identified system limitations as mentioned in SPP Planning Criteria section 6.2.4.
- 2) Line Outage Distribution Factor - The Line Outage Distribution Factor (LODF) is the percent of the power flowing across the contingency facility that transfers over the monitored facility when the contingency facility is switched out of service.
- 3) Power Transfer Distribution Factor - The Power Transfer Distribution Factor (PTDF) is the percentage of a power transfer that flows through a facility or a set of facilities for a particular transfer when there are no contingencies. PTDF type Flowgates are used for representing Thermal, Voltage, Stability and Contractual Limitations. To be considered a valid limit to transfers, a PTDF Flowgate must have a PTDF at or above the applicable short-term or long-term threshold.
- 4) Outage Transfer Distribution Factor - The Outage Transfer Distribution Factor (OTDF) is the percentage of a power transfer that flows through the monitored facility for a particular transfer when the contingency facility is switched out of service. OTDF type Flowgates typically represent contingency based thermal limitations within the system. They can also be used to represent Stability limitations. To be considered a valid limit to transfers, a Monitored Facility must have an OTDF at or above the applicable short-term or long-term threshold.
- 5) Power Transfer Voltage Factor - The Power Transfer Voltage Factor (PTVF) is the per unit amount that a facility's voltage changes due to a particular transfer level. To be considered a valid limit to transfers, a Monitored Facility must have a PTVF at or above the applicable short-term or long-term threshold.

6.2.4 **TRANSFER CAPABILITY LIMITATIONS**

The electrical ability of the interconnected transmission network to reliably transfer electric power may be limited by any one or more of the following:

- 1) Thermal Limits - Thermal limits establish the maximum amount of electrical current that a transmission circuit or electrical facility can conduct over a specified time period before it sustains permanent damage by overheating or before it violates public safety requirements. Normal and emergency transmission circuit ratings are defined in the SPP Rating of Equipment.
- 2) Voltage Limits - System voltages must be maintained within the range of acceptable minimum and maximum voltage limits. For example, minimum voltage limits can establish the maximum amount of electric power that can be transferred without

causing damage to the electric system or customer facilities. A widespread collapse of system voltage can result in a blackout of portions of or the entire interconnected network. Acceptable minimum and maximum voltages are network and system dependent. The Normal Voltage Limit range is the operating voltage range on the interconnected systems, above or below nominal voltage and generally expressed in kilovolts that is acceptable on a sustained basis. The Emergency Voltage Limit range is the operating voltage range on the interconnected systems, above or below nominal voltage and generally expressed in kilovolts that is acceptable for the time sufficient for system adjustments to be made following a facility outage or system disturbance. Voltage limits will be as specified in the SPP Planning Criteria section 5.

- 3) Stability Limits - The transmission network must be capable of surviving disturbances through the transient and dynamic time periods following a disturbance. Specific Stability Limits Criteria is found in the SPP Criteria: Regional Transmission Coordinated Planning.
- 4) Contractual Requirements - Some Transmission Owners have contractual arrangements that contain mutual agreements regarding the power transfer available between them. These contractual arrangements have been approved by the appropriate regulatory agencies. The NERC Operating Policies inherently recognize contract requirements that may limit the power transfer between Transmission Owners. Some contract requirements are discussed in NERC Operating Policy 3 – Interchange.

The limiting conditions on some portions of the transmission network can shift among thermal, voltage, stability and contractual limits as the network operating conditions change over time.

6.2.5 **INVALID LIMITS**

The procedures outlined in criteria may lead to identification of certain limiting facilities that are invalid. Reasons may include, but are not limited to:

- 1) An invalid contingency generated as a generic single outage, which is not valid without the outage of other facilities.
- 2) Incorrect ratings. Ratings will be corrected and the limiting transfer level recalculated.
- 3) The rating used may be directional in nature (directional relaying) and may not be valid for the direction of flow.
- 4) The limiting facility is the result of over-generation/under-generation at a participation point.
- 5) The contingency is considered improper implementation of an operating procedure.
- 6) The facility represents an equivalent circuit.
- 7) The limiting facility is declared a Local Area Problem.

Any limiting facility determined to be invalid due to modeling error that could be corrected must be corrected by the next series of seasonal calculations.

6.2.6 **FLOWGATES**

Flowgates are selected power transmission element groups that act as proxies for the power transmission system capable of representing potential thermal, voltage, stability and contractual system limits to power transfer. There are two types of Flowgates;

- (1) OTDF Flowgate - Composed of usually two power transmission elements in which the loss of one (contingency facility) can cause the other power transmission element (monitored facility) to reach its emergency rating.
- (2) PTDF Flowgate - Composed of one or more power transmission elements in which the total pre-contingency flow over the flowgate cannot exceed a predetermined limit. Either with the power transmission system intact or with a contingency elsewhere, the Flowgate can be selected to represent a thermal, voltage, stability or contractual limit.

Once a set of limiting elements have been identified, as potential transfer constraints, they can be grouped with their related components and identified as unique Flowgates. The rating of the Flowgate is called the Total Flowgate Capacity (TFC) of the Flowgate and is monitored and used for evaluation of all viable transfers for commerce. To the extent that the impedance network models are similar with similar participation patterns, the same Flowgates can be monitored in other network models for purposes of evaluating the impact of additional transactions on the network. Of course, each network model will be subtly different therefore it is important that engineering judgment is exercised regarding the validity of applying existing Flowgates to a new network model.

6.2.7 **TOTAL FLOWGATE CAPACITY (TFC)**

The Flowgate and its Total Flowgate Capacity are pre-defined. A Flowgate is intended to limit the amount of power allowed to flow over a defined element set. This TFC may reflect several possible types of system limitations as described in SPP Planning Criteria section 6.2.4.

For OTDF Flowgates representing thermal overloads, the TFC represents the total amount of power that can flow during the contingency without violating the emergency rating of the monitored facility.

For PTDF Flowgates, the TFC represents the total amount of power that can flow over a defined element set under pre-contingency conditions.

Again, limit types could be:

- 1) Thermal limits under normal operating conditions or linked contingency events,
- 2) Voltage limits under normal operating conditions or linked contingency events,
- 3) Stability limits under normal operating conditions or linked contingency events, or
- 4) Contractual limits.

Flowgates are selected based on the impacts of power transfer in an electrical network and will be evaluated on a regular basis and revised as needed to ensure thorough representation of the system they are representing. Each Flowgate represents a possible limitation within a network and in itself has a Flowgate rating (TFC) and an Available Flowgate Capacity (AFC)

which can be translated via the path response factor (TDF) to a path Available Transfer Capability (path ATC) for any path.

6.2.8 **FLOWGATE CAPACITY**

6.2.8.1 ***Total Flowgate Capacity (TFC)***

A Flowgate acts as proxy to path transfer limitations. This allows additional transfer capability on a path based on the additional loading that can be incurred. The determination of additional loading that can be incurred on a Flowgate begins first with the determination of the maximum loading that can be allowed on a PTDF Flowgate or on the monitored facility of an OTDF Flowgate during its associated contingency. This maximum loading is termed Total Flowgate Capacity (TFC).

6.2.8.2 ***Available Flowgate Capacity (AFC)***

The available capacity on a Flowgate for additional loading for new power transfers is determined by taking the Total Flowgate Capacity (TFC) and removing the Flowgate Base Loading (FBL) and the Impacts due to existing system commitments and any transmission margins.

$AFC = TFC - FBL - \text{Impacts of existing commitments} - \text{transmission margins}$

6.2.8.3 ***Firm and Non-Firm Available Flowgate Capacity (FAFC and NFAFC)***

Path ATC is classified as firm or non-firm. This distinction is made when determining the AFC remaining for path ATC. AFC is classified as firm or on firm depending on the types of existing commitments considered for Impacts. This is realized in the formula for AFC:

$AFC = TFC - FBL - \text{Impacts of existing commitments} - \text{transmission margins}$

6.2.9 **SYSTEM IMPACTS**

6.2.9.1 ***Impacts of Existing Commitments***

In order to simultaneously account for impacts of all commitments to all paths at any given instant in time, it is necessary to devise a system that allows for fluctuation in the number of and the magnitude of system commitments on each path within an acceptable amount of time, for the purpose of providing transmission service in a competitive manner.

Existing transmission commitments beyond those modeled as native load and related generation commitments can be found on the OASIS. However, before impacts of OASIS posted reservations can be calculated, they must first be interpreted – carefully examined for peculiar individual characteristics. Due to the nature of the OASIS and the rules therein, posted reservations sometimes require interpretation as to their actual value to apply toward the transmission network.

The following are examples of evaluations that are performed:

- 1) Recognize and adjust for duplicate reservations by multiple providers to complete one transaction.
- 2) Adjust for reservations that may have changed status or have been replaced by another reservation, including renewals and redirects.
- 3) Check for proper reflection of capacity profiles of reservations.
- 4) Distinguish status and class of reservations such as Study, Accepted, Confirmed, Firm, and Non-Firm status to determine their proper impact level.

6.2.9.2 Positive Impacts

The scope of “Impacts of existing commitments” in the formula for AFC incorporates both the calculated positive impacts and counter impacts of non-firm and firm service commitments. A positive impact is determined as having the effect of increasing the loading on a Flowgate in the direction of the Flowgate. Positive impact types are sorted into those resulting from firm and non-firm service commitments. To determine firm or non-firm Available Flowgate Capacity, the appropriate impacts are applied to make up the “Impacts of existing commitments” in the above formula. Additionally, counter impacts are considered depending on firm or non-firm determinations.

6.2.9.3 Counter Impacts

Counter impacts are those impacts due to transfers that act to relieve loading on limiting elements. Counter impact types are sorted into those resulting from firm and non-firm service commitments. These flows are not traditionally accepted as valid under the pretense that any reservation that may cause such a loading relief is not actually doing so until it has been scheduled. To consider counter-flows in transfer capability studies is to assume a high probability of scheduling.

6.2.10 MONITORED FACILITIES

During the Flowgate determination process those facilities monitored for pre-defined limiting conditions. Mandatory Monitored Facilities, for use in these calculations, are all facilities operated at 100 kV and above and all interconnections between Transmission Providers. Other facilities operated at lower voltage levels may be added to the Monitored Facilities list at the discretion of the Transmission Providers or Transmission Owners. In defining Flowgates, the Monitored Facilities are those components of a Flowgate that remain in service following the defined contingency.

6.2.11 CRITICAL CONTINGENCIES

Those facilities that, when outaged, are deemed to have an adverse impact on the reliability of the transmission network. These facilities may be transmission facilities, including multi-terminal lines, or generating units. All interconnections of an area will be considered Critical Contingencies, regardless of voltage level as will the largest generating unit in the area.

6.3 RELIABILITY MARGINS

Transmission margins are very important to the reliability of the interconnected network in an Open Access environment. The NERC "Available Transfer Capability Definitions and Determination Reference Document" defines Transmission Reliability and Capacity Benefit margins (TRM, CBM).

When using Flowgates as a means to represent a system's constraints, it is necessary to translate reliability margins, TRM and CBM, to a unique TRM and CBM for each Flowgate. Margins are the required capacities that must be preserved for the purpose of moving power between areas during specific emergency conditions. Since a margin is a preservation of transfer capacity, the margin itself will have an impact on the most limiting element between the two areas for which it is reserved.

All studies for the purpose of assessing TRM and CBM will only include generation units located within the transmission system for which the Transmission Provider is responsible. These generation units may also include those not specifically designated to serve network load connected to transmission systems within the Transmission Provider system. However, the method by which a Transmission Provider is to determine TRM and CBM shall not vary from that described herein with the exception of assessing facilities located outside of SPP regional structure/bounds.

6.3.1 TRANSMISSION RELIABILITY MARGIN (TRM)

TRM on a Flowgate basis is that amount of reserved Flowgate capacity necessary to ensure that the interconnected transmission network is secure under a reasonable range of uncertainties in system conditions. The following factors shall be considered by SPP in the determination of TRM:

- 1) Load Forecast: Transmission Providers will forecast hourly load for the next seven days for all applicable control areas. Beyond seven days, Transmission Providers will project a demand based on seasonal peak load models for all applicable Transmission Owners. These load levels will be the projected peaks for the time frame for which the forecast applies.
- 2) Variations in Generation Dispatch: Variations to generation patterns constitute a viable concern. Generation dispatch in near-term models will be based on real-time snapshots of network system conditions. For the longer-term horizons, whenever possible, generation dispatch information provided by generation owners will be applied to the ATC calculations. However, it is recognized that longer-term dispatch is probably unknown to the generation controlling entities themselves except for base-load and must run type units.
- 3) Unaccounted Parallel Flows: Parallel flows can be an issue if pertinent data to the ATC calculations are not shared among the transmission providers and those transactions that have multiple wheeling parties are not identified. Provisions in the SPP OATT have reduced the impacts of these transactions within SPP and between SPP and other regions. Transmission Owners of facilities that are impacted by unaccounted parallel flows or variations in dispatch may request additional TRM for their impacted

Flowgates from the TWG. Such requests must be in writing, must document the parallel flow impacts or the variance in historical dispatch, and be accompanied by analysis or documentation supporting the additional TRM requirements. The TWG shall have the authority to grant or reject requests for the additional TRM requests.

- 4) SPP Operating Reserve Sharing: The SPP Operating Reserve Sharing program was instituted to provide both reliability and economic benefits to its members. This program reduces the amount of internal operating reserves each entity is required to maintain while providing an automated way of allocating resources on a region wide level to ensure quick recovery for the loss of any unit. Transmission facilities must be able to support the automatic implementation of the Reserve Sharing program. To that end, TRM on the Flowgates will provide enough capacity to withstand the impact of the most critical generation loss to that facility. All generation contingencies will be simulated by the Operating Reserve Sharing algorithm to determine the highest impact on each Flowgate. This capacity will be included in TRM.
- 5) Counter Flow Impacts: Another factor to consider in the SPP TRM process is that for the planning horizon, which is primarily next day and beyond, the counter flow impacts of reservations on the Flowgates are removed with the exception of Designated Network Resources. This provides an inherent margin in the calculation which along with the constant TRM provided by the reserve sharing allocation, is a proxy for the generation variation.

6.3.2 **TRM COORDINATION**

The TRM specified on a Flowgate represents a transmission margin that the transmission system needs to maintain a secure network under a reasonable range of uncertainties in system conditions. As such it is not necessarily an import or export quantity specifically. The Automatic Operating Reserve Sharing portion is determined by centralized Regional study based on the SPP Operating Reserve Sharing Criteria. Any additional TRM may be requested by the Flowgate owner(s), subject to review by the SPP TWG.

6.3.3 **TRM AVAILABILITY FOR NON-FIRM SERVICE**

To maximize transmission use to the extent reliably possible, Transmission Providers may sell TRM on a non-firm basis. The 'a' and 'b' multipliers facilitate this purpose in the calculations. However, a contingency or long-term outage to a high impact unit may result in the curtailment of non-firm schedules and displacement of non-firm reservations sold within the TRM.

6.3.4 **TRM CALCULATION FREQUENCY**

The Operating Reserve Sharing portion of the TRM will be determined annually for each season (spring, summer, fall, winter). This process is outlined in the SPP Criteria under Operating Reserves and the Operating Reserve Share Program Procedures. Flowgate owner requests for additional TRM may be submitted at any time for consideration at the next TWG meeting. The submittal should include justification and rationale in writing for the requested additional TRM. The TWG shall have authority to reject or grant such requests.

6.3.5 **CAPACITY BENEFIT MARGIN (CBM)**

CBM on a Flowgate basis is the amount of Flowgate capacity reserved by load serving entities to ensure access to generation from interconnected systems to meet generation reliability requirements.

SPP does not utilize CBM for calculations of ATC for some or all of the following reasons:

- 1) the existing level of internal reserve margin of each member is adequate
- 2) historical reliability indicators of transmission strength of the SPP area
- 3) Open Access transmission usage environment allows greater purchasing options

Since SPP does not utilize CBM for any flowgate within the SPP footprint, the CBM value used in any calculations will be zero.

7. ELECTRICAL FACILITY RATINGS

7.1 ACCREDITATION AND TESTING REQUIREMENTS

This section shall be used separately to determine the ~~annual and~~ seasonal Accredited Capacity and Net Generating Capability of resources and programs for the SPP East BAA and SPP West BAA. Procedures are herein for establishing a system of records so that changes in capacity during the life of the equipment can be recognized. The terms defined and the capacity established pursuant to these procedures shall be used for the Transmission Provider purposes, including determining planning reserve margins for resource adequacy. These capacity requirements are not intended to restrict daily operating practices associated with Transmission Provider's Markets and Operations. Each member shall test and accredit its resource's equipment in accordance with the procedures contained herein. On the basis of these tests, seasonal capability ratings for each resource shall be established. For Behind-The-Meter Generation, section 7.1 applies only to Controllable and Dispatchable Behind-The-Meter Generation, as defined in Attachment AA of the SPP OATT. Non-Controllable and Non-Dispatchable Behind-The-Meter Generation, as defined in Attachment AA of the SPP OATT, do not require testing. For any reference of "demonstrated net generating capability" in Attachment AA of the SPP OATT shall be referred to as "Net Generating Capability" for ~~units-resources~~ internal to the applicable SPP Balancing Authority Area and applicable to this criteria section 7.1.

7.1.1 DEFINED TERMS

- 1) Rated Ambient Temperature: The station dry-bulb (DB) Summer Season rated ambient temperature is from the American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE) Fundamentals Handbook, Climatic Design Information, Cooling and Dehumidification Design Conditions - Cooling dry-bulb (DB)/mean coincident wet-bulb (MCWB) 0.4%. If the resource is within 30 miles of the nearest weather station reported in the ASHRAE Fundamentals Handbook, then these temperatures will be those for the nearest station. For all other stations, rating temperatures shall be determined by interpolating between weather stations using plant latitude and longitude. The steps to be used for interpolating weather data and correcting for elevation are presented in the SPP Planning Criteria, section 9.
- 2) Capability Test: The demonstration of capability of certain resources or programs by generating at their rated capability under specified test conditions and test duration.
- 3) Operational Test: The annual demonstration of the functional ability of a resource or program.
- 4) Summer Season: ~~The Summer Season is defined as the months of June, July, August, and September (June 1-September 30)As defined in Attachment AA of the SPP OATT.~~
- 5) Winter Season: ~~The Winter Season is defined as the months of December, January, February, and March (December 1-March 31)As defined in Attachment AA of the SPP OATT.~~
- 6) Net Generating Capability: ~~As defined in Attachment AA of the SPP OATTThe gross maximum output a unit can sustain over a one-hour period reduced by any power used for unit auxiliaries, modified for environmental, seasonal, operational and fuel limitations.~~

Commented [SP5]: RR703 Demand Response

- Approved by the RSC on:
- Approved by the Board on:
- Posted with FERC: 4/17/2026
 - FERC Docket #: ER26-2249-000
- Requested FERC Effective Date: 1/1/2027

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- Approved by the Board on:
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- Requested FERC Effective Date: 1/1/2027

- 7) Accredited Capacity: As defined in Attachment AA of the SPP OATT.
- 8) Net Head: The water column height that represents the hydraulic power available to a hydroelectric generating unit to derive electrical energy, calculated as the difference between headwater elevation and tailwater elevation.
- 9) Net Peak Load: As defined in Attachment AA of the SPP OATT.
- 10) Energy Storage Resource: As defined in Part I, Section 1 of the SPP OATT.
- 11) Demand Response Capacity: As defined in Attachment AA of the SPP OATT.
- 12) Market Participant: As defined in Attachment AE of the SPP OATT.
- 13) Demand Response Resources: As defined in Attachment AE of the SPP OATT.
- 14) Conservative Operations Demand Response Programs: As defined in Attachment AA of the SPP OATT.
- 15) Emergency Demand Response Programs: As defined in Attachment AA of the SPP OATT.
- 14)16) Reliability Registered Demand Response Programs (RRDRP): As defined in Attachment AE of the SPP OATT.

7.1.1.2 **CAPABILITY TEST**

Capability Tests establish the Net Generating Capability of a unit. If a Capability Test is unable to be performed, the out of season testing procedure must be followed. Generation specific guidelines are provided. Test data shall be compiled and submitted via the Workbook submittal process as outlined in Attachment AA of the SPP OATT. The Summer Season Net Generating Capability may be used for the Winter Season without further testing. Run of River Hydro, Wind, and Solar units are excluded from Capability Testing.

7.1.2.1 **Thermal Units**

Thermal units include traditional resources that utilize thermal energy to produce electricity via a generator either directly (i.e. combustion turbine) or indirectly via an intermediate steam cycle (steam turbine). Examples of thermal units include steam boilers, combustion turbines, combined cycle, nuclear units, reciprocating engines, etc.

- 1) The Capability Test shall be performed every five years (not to exceed 64 months) during the Summer Season. If anytime during the five years a new Capability Test is submitted to the Transmission Provider, the five-year window starts over. If the unit has a higher claimed Winter Season capability, a separate Winter Season Capability Test shall also be performed during the Winter Season. Test duration shall be a minimum of one hour.
- 2) Summer Season Capability Tests are to be conducted during a time when ambient DB temperature is no more than 10°F below the station ASHRAE Rated Ambient Temperature. At the time of testing, the most recent version of the ASHRAE Fundamentals Handbook shall be utilized. If the DB temperature exceeds 10° below the ASHRAE rated temperature, a penalty of 5% plus an additional 0.5% per degree for each additional degree below 10°, up to 20°, will be applied to the Capability Test result. A Summer Season Capability Test shall not be performed in excess of 20° the ASHRAE rated temperature. There is no ambient temperature requirement for Winter Season Capability Tests.

Commented [SP8]: RR703 Demand Response

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- Posted with FERC: 4/17/2026
 - FERC Docket #: ER26-2249-000
- Requested FERC Effective Date: 1/1/2027

- 3) The unit shall be brought to the desired test load and allowed to stabilize. Once the test period has begun, only minor changes in unit controls shall be made as required to maintain the unit in normal, steady-state operation.
- 4) The unit capability shall be determined separately for each generating unit in a power plant where the input to the prime mover of the unit is independent of the others.
- 5) Units that are aggregated into a commercial resource registration and prefer testing aligned with their registered resource and/or dependent upon common systems (i.e. fuel, steam supply, aux equipment, transmission, etc.) which restrict total output shall be tested simultaneously. Each unit shall be assigned an individual capability by apportioning the combined capability among the units.
- 6) The fuel used during testing shall be the type expected to be used during peak load conditions.
- 7) The Net Generating Capability of a unit or plant obtained through non-typical operation (i.e. bypassing feedwater heaters, varying steam conditions, alternate control mode, etc.) is acceptable if the following conditions are met:
 - (a) The capability based on such conditions shall be available for a period of not less than four continuous hours.
 - (b) The appropriate procedures have been established so that this capability is available for four continuous hours.
- 8) The Summer Season or Winter Season Net Generating Capability established for Behind-The-Meter Generation, which does not have firm delivery beyond a discrete point of delivery, shall be adjusted in accordance with Behind-The-Meter Generation Reporting and Documentation Procedures given in the SPP Business Practices.

7.1.2.2 Conventional & Pumped Storage Hydro Unit

- 1) Summer Season Capability Tests are to be conducted during a time when the Net Head is no more than 10% above the seasonal median Net Head recorded for the hydroelectric plant. The seasonal median Net Head will be determined from the most recent 10 years of historical hourly data for each month of the season evaluated. Facilities in commercial operation three years or less must include the most recent three years of the seasonal median Net Head. Facilities in commercial operation four years and greater must include all available data up to the most recent 10 years of the seasonal median Net Head.
 - (a) If the Net Head at time of the Capability Test exceeds 10% above the seasonal median Net Head, then the hydroelectric plant Capability Test value shall be the lower of:
 - i. The actual Capability Test result
 - ii. The Capability Test result corrected to the plant capability at 110% of the seasonal median Net Head.
- 2) For new hydro units, the unit specific engineering study may be utilized in place of the Net Head. After the first year of operation, the hydro unit must provide the seasonal Net Head.
- 3) The Capability Test shall be performed every five years (not to exceed 64 months) during the Summer Season. If anytime during the five years a new Capability Test is submitted to the Transmission Provider, the five-year window starts over. Test duration shall be a minimum of one hour.

- 4) For units that utilize a Winter Season Net Generating Capability higher than Summer Season Net Generating Capability, Capability Tests shall be performed every five years (not to exceed 64 months) during the Winter Season.

7.1.2.3 Energy Storage Units

- 1) Energy Storage Resources (ESR) must be tested to the time (e.g. four-hour) duration at which they are accredited. The operational performance of the energy storage resource may be used to satisfy the requirement of the Capability Test.
- 2) The Capability Test shall be performed every three years during the Summer Season.
- 3) For units that utilize a Winter Season Net Generating Capability higher than the Summer Season Net Generating Capability, Capability Tests shall be performed every three years during the Winter Season.

7.1.2.4 Out of Season Capability Test

If the unit is due for a Capability Test and the unit cannot perform the Capability Test due to a forced outage, a maintenance outage, or a forced de-rate, the most current Capability Test results may be used for the upcoming Summer Season only. The unit will be required to perform an Operational Test following the Operational Test procedures before the next Summer Season. For example, a unit is entered a forced outage while performing a Capability Test. The repair for the unit cannot be completed until after the Summer Season. When the unit is repaired, an Operational Test must be completed. The previous Capability Test will be used to satisfy Attachment AA of the SPP OATT for the Workbook submittal for the upcoming Summer Season. A Capability Test must be performed in the next Summer Season for the next Workbook submittal. If the unit fails to complete a Capability Test, the unit cannot be claimed on the Workbook submittal.

7.1.3 OPERATIONAL TEST

An Operational Test is used as an annual demonstration of the functional ability of a resource. This test must be completed in a 12-month period beginning on May 15 of each year and can be conducted in any season. Test data shall be compiled and submitted via the Workbook submittal process as outlined in Attachment AA of the SPP OATT. If a unit has both Summer Season and Winter Season Net Generating Capability, the Operational Test must be conducted at a minimum of 90% of the Summer Season Net Generating Capability. The Operational Test shall be conducted for a minimum of one hour and there are no Rated Ambient Temperature requirements for Operational Tests.

7.1.3.1 Thermal, Conventional & Pumped Storage Hydro Units

Operational Tests shall be conducted at a minimum of 90% of current Summer Season Net Generating Capability. Any hour with the unit operating at or above 90% of Net Generating Capability may be deemed an Operational Test. In case of failure to meet 90% of Net Generating Capability, the unit can only claim what it can achieve on the Operational Test on the upcoming Workbook submittal. A new Capability Test will need to be performed to re-establish the Net Generating Capability of the unit.

7.1.3.2 Run of River Hydro Units

Operational Tests shall be conducted at a minimum of 90% of current Summer Season Accredited Capacity. Any hour with the unit operating at or above 90% of the Accredited Capacity may be deemed an Operational Test. In case of failure to meet 90% of the Accredited Capacity, the unit can only claim what it can achieve on the Operational Test on the upcoming Workbook submittal.

7.1.3.3 Wind and Solar Units

Operational Tests shall be conducted at a minimum of 100% of the current Summer Season Accredited Capacity. Any hour with the unit operating at or above 100% of the Accredited Capacity may be deemed an Operational Test. In case of failure to meet 100% of the Accredited Capacity, the unit can only claim what it can achieve on the Operational Test upcoming Workbook submittal.

7.1.3.4 Energy Storage Units

Operational Tests shall be conducted at a minimum of 100% of the current Summer Season Accredited Capacity. Any hour with the unit operating at or above 100% of the Accredited Capacity may be deemed an Operational Test. In case of failure to meet 100% of the Accredited Capacity, the unit can only claim what it can achieve on the Operational Test on the upcoming Workbook submittal.

7.1.4 NEW OR UPGRADED UNIT TESTING

For newly installed units and units undergoing a physical or operational modification which could impact a unit's Net Generating Capability, design output may be used for the first peak operating season to allow sufficient time for Operational and Capability Tests to be conducted. A Capability Test shall be performed in the upcoming peak Summer Season in order to establish the new Net Generating Capability going forward beyond the first peak Summer Season.

7.1.5 ACCREDITATION FOR THERMAL AND HYDRO UNITS

Assurance of having desired generating capacity depends, in part, on the availability of an adequate and reliable fuel supply. Where contractual or physical arrangements permit curtailment or interruption of the normal fuel supply, sufficient quantities of standby fuel shall be provided. Due to the dependence of hydroelectric plants on seasonal water flows, this factor shall be taken into consideration when calculating the Net Generating Capability.

7.1.5.1 Thermal, Conventional & Pumped Storage Hydro Units

Ambient condition load corrections are not required but may be applied at the member's discretion. Correction methodology must be documented and maintained available for review by the Transmission Provider's staff. The Net Generating capability shall be determined taking into consideration the fuel management program and any restrictions imposed by regulatory agencies. The Accredited Capacity for thermal, conventional hydro and pumped storage hydro units shall be established in accordance with Attachment AA of the SPP OATT and SPP Business Practices.

7.1.5.2 Run of River Hydro Units

The recommended methodology to evaluate the Accredited Capacity established for run of river hydro facilities shall be determined on a seasonal basis, as stated below. If a member desires to use a more restrictive methodology to evaluate the Accredited Capacity of a run of river hydro facility, they may do so, however Accredited Capacity determined by the alternative methodology employed cannot credit the facility with a capability greater than determined with the methodology stated below:

- 1) Assemble all available hourly net power output (MWh) data measured at the system interconnection point.
 - (a) Select the hourly median net power output values occurring during the top 3% of load hours for the entity for each month of each year for the evaluation period.
 - (b) A seasonal Accredited Capacity may be determined by selecting the appropriate monthly MW values corresponding to the Load Responsible Entity's peak load month of the season of interest.
 - (c) The data must include all available data with a minimum of five years up to the most recent 10 years of commercial operation.
 - (d) Only metered hourly net power output (MWh) data may be used.
 - (e) If the entity does not perform or provide the Accredited Capacity calculations to the Transmission Provider as described above, then the Accredited Capacity for the unit will be zero MW.
- 2) The Accredited Capacity calculation shall be updated at least once every three years.

7.1.6 EFFECTIVE LOAD CARRYING CAPABILITY (ELCC) FOR SOLAR AND WIND UNITS

Effective June 1, 2026, for the SPP East BAA's 2026 Summer Season, Accredited Capacity for wind and solar resources will be established using ELCC methodology. Effective June 1, 2027, for the SPP West BAA's Summer Season, Accredited Capacity for wind and solar resources will be established using ELCC methodology. As outlined in Attachment AA of the SPP OATT, SPP will utilize historical output to analyze and allocate the total system Accredited Capacity to each wind and solar facility. The ELCC study for wind and solar resources will be performed separately for the SPP East BAA and the SPP West BAA once a year for the Summer and Winter Seasons, and the results can be applied for the upcoming planning years in the annual Workbook submissions for the appropriate SPP BAA, as outlined in Attachment AA of the SPP OATT.

7.1.6.1 Tier Determination

Wind and solar units will be studied in tiers on meeting the requirements outlined in Attachment AA of the SPP OATT.

7.1.6.2 Allocation

Once the total Accredited Capacity value has been determined for each tier through the ELCC Study process, each resource will be assigned a percentage of the total ELCC MW from its corresponding tier in accordance with Section 15 of Attachment AA.

The load data used in all tiers will include values for the most recent three years for each resource's average production calculation. Each LRE is required to annually submit their previous years hourly load data to the Transmission Provider. If the LRE does not provide the data by June 1, the LRE's resource(s) will be studied and allocated with resources in Tier 2.

The total ELCC MW determined in all tiers will be fully allocated through this process. As necessary, single facilities that are divided into multiple tiers will have the associated LRE or SPP Balancing Authority Area's load shapes applied to that portion of the resource. Curtailments based on operational market data, for wind and solar resources will be added back into the historical production output, as available. In accordance with the SPP Business Practices and Attachment AA of the SPP OATT, Out of Management Control ("OMC") events, as defined by NERC, that impact the average production calculation of an individual wind and solar resources shall be excluded from the determination of allocation of such resources. The historical output used for each resource in the allocation process is represented in two categories, based on the commercial operation date, listed below.

- 1) Resources that have been in commercial operation for less than three years:
 - (a) A new wind or solar resource that does not have three years of operational data may provide on-site weather data (such as wind speed and solar irradiance data), and facility attributes (such as turbine power curves, inverter load ratios, etc.) to create a power production estimate by the resource owner or operator to produce an engineered shape. Wind data correlated with a reference tower beyond fifty miles and solar data correlated with a reference-measuring device beyond 200 miles, is subject to Supply Adequacy Working Group approval.
 - (b) If a new wind or solar resource does not provide engineered shapes based on historical data, the Transmission Provider will generate output data using the metered hourly net power output (MWH) data from the nearest wind or solar resource with a comparable capacity factor or technology vintage to complete the appropriate data set. If no nearby resource has a comparable capacity factor, the output data from the nearest existing resource will be scaled up to mimic the power curve for the technology of the new resource.
 - (c) If a new wind or solar resource does not provide historical estimated output that facility will be included in the annual ELCC study as a new facility in accordance with section 7.1.6.3.
- 2) Resources that have been in commercial operation greater than three years:
 - (a) The Transmission Provider will consider, at a minimum, the most recent three years of data when selecting the top 3% Net Peak Load hours, up to no more than the most recent seven years of commercial operation.
 - (b) Only metered hourly net power output (MWH) data at the point of interconnection will be used.

If a wind or solar resource that has been in commercial operation greater than three years undergoes a technology change, the Transmission Provider will continue to use the previous years' electrical output as specified in, section 7.1.6.2, subsection (1) above, unless the LRE or Generation Owner chooses to treat the existing resource as a new resource. If treated as a new resource it would be considered as a wind or solar resource that has been in commercial

operation three years or less. The LRE or Generation Owner must notify the Transmission Provider no later than June 1 of for the Summer Season ELCC Study and December 1 for the Winter Season ELCC Study. Once a new date is established, section 7.1.6.2, subsection (1) above would apply to the resource if the new commercial operation date is less than three years.

7.1.6.3 New Wind and Solar Units

7.1.6.3.1 Summer Season

For new wind or solar resources that reach commercial operation after June 1 of the applicable study year, the resources must be submitted by the appropriate LRE or Generator Owner in the applicable SPP BAA Workbook by May 15 to be studied in the upcoming ELCC study for that year. If submitted after May 15, the facility would receive a 10% accreditation for wind and a 50% accreditation for solar for the following Summer Season after the May 15 deadline or the average of the lowest tier whichever is lower.

- 1) The following must be completed to have a new resource be included in Tier 1 of the applicable ELCC study year:
 - a) Resource submitted in the Generator Owner or LRE's current year's Workbook by May 15.
 - b) LRE or Generator Owner provides three years of engineered shapes with the first year being the same as the current submittal year. If a new wind or solar resource does not provide engineered, the Transmission Provider will follow procedures outlined in, section 7.1.6.2, subsection (1)(b).
 - c) Designated Resources, resources associated with Point-to-Point Transmission Service, and other resources with firm transmission service must confirm the appropriate level of service by June 1 of the current year that spans June 1 to September 30 for the Summer Season of the following year.
- 2) The following must be completed to have a new resource be included in Tier 2 of the applicable ELCC study year:
 - a) Resource submitted in the Generator Owner or LRE's current year's Workbook by May 15.
 - b) LRE or Generator Owner provides three years of engineered shapes with the first year being the same as the current submittal year. If a new wind or solar resource does not provide engineered shapes, the Transmission Provider will follow procedures outlined in, section 7.1.6.2, subsection (1)(b).

7.1.6.3.2 Winter Season

For new wind or solar resources that reach commercial operation after December 1 of the applicable study year, the resources must be submitted by the appropriate LRE or Generator Owner in the applicable SPP BAA Workbook by November 15 to be studied in the upcoming ELCC study for that year. If submitted after November 15, the facility would receive a 10% accreditation for wind and a 50% accreditation for solar for the following Winter Season after the November 15 deadline or the average of the lowest tier whichever is lower.

- 1) The following must be completed to have a new resource be included in Tier 1 of the applicable ELCC study year:

- a) Resource submitted in the Generator Owner or LRE's current year's Workbook by November 15.
 - b) LRE or Generator Owner provides three years of engineered shapes with the first year being the same as the current submittal year. If a new wind or solar facility does not provide engineered shapes, the Transmission Provider will follow procedures outlined in, section 7.1.6.2, subsection (1)(b).
 - c) Designated Resources, resources associated with Point-to-Point Transmission Service, and other resources with firm transmission service must confirm the appropriate level of service by December 1 of the current year that spans December 1 to March 31 for the Winter Season of the following year.
- 2) The following must be completed to have a new resource be included in Tier 2 of the applicable ELCC study year:
- a) Resource submitted in the Generator Owner or LRE's current year's Workbook by November 15.
 - b) LRE or Generator Owner provides three years of engineered shapes with the first year being the same as the current submittal year. If a new wind or solar resource does not provide engineered shapes, the Transmission Provider will follow procedures outlined in, section 7.1.6.2, subsection (1)(b).

7.1.7 ELCC ACCREDITATION FOR ENERGY STORAGE UNITS

Effective June 1, 2026, for the SPP East BAA's 2026 Summer Season, Accredited Capacity for an ESR, excluding pumped storage hydroelectric resources, will be established using ELCC methodology as outlined in Attachment AA of the SPP OATT. Effective June 1, 2027, for the SPP West BAA's 2027 Summer Season, Accredited Capacity for an ESR, excluding pumped storage hydroelectric resources, will be established using ELCC methodology as outlined in Attachment AA of the SPP OATT. The total ELCC MW will be determined by the annual ELCC study process for ESRs with four hour, six hour, and eight hour durations for each SPP East BAA and SPP West BAA.

7.1.7.1 New Energy Storage Units

7.1.7.1.1 Summer Season

For new ESRs that reach commercial operation after June 1 of the applicable study year, the resources must be submitted in the Workbook by May 15 to be studied in the upcoming ELCC study. For newly submitted resources that were not studied in the previous ELCC study:

- 1) In year one, Tier 2 accreditation percentage from the latest ELCC curve will be assigned to the resource based on the appropriate amount of nameplate penetration in accordance with Attachment AA of the SPP OATT.
- 2) In year two of service, before the next biennial ELCC study is performed, if the resource qualifies for Tier 1 by May 15, the accreditation percentage from the latest ELCC curve will be applied based on Tier 1 results on the appropriate amount of nameplate penetration in accordance with Attachment AA of the SPP OATT.

7.1.7.1.2 Winter Season

For new ESRs that reach commercial operation after December 1 of the applicable study year, the resources must be submitted in the Workbook by November 15 to be studied in the upcoming ELCC study. For newly submitted resources that were not studied in the previous ELCC study:

- 1) In year one, Tier 2 accreditation percentage from the latest ELCC curve will be assigned to the resource based on the appropriate amount of nameplate penetration in accordance with Attachment AA of the SPP OATT.
- 2) In year two of service, before the next biennial ELCC study is performed, if the resource qualifies for Tier 1 by November 15, the accreditation percentage from the latest ELCC curve will be applied based on Tier 1 results on the appropriate amount of nameplate penetration in accordance with Attachment AA of the SPP OATT.

7.1.7.2 Establishing Energy Storage Resource ELCC Curves

Biennially, the Transmission Provider will perform an ELCC study for future planned ESR installations to generate the ELCC curves for each duration type for each SPP BAA, as outlined in Attachment AA of the SPP OATT. The future ESR installations will be studied in block increments to be determined in the scope of the study. Each block increment will be composed of the three types of duration devices (4 hour, 6 hour, and 8 hour). Upon completion in determining the ELCC curve for each duration type for each SPP BAA, the Transmission Provider will post the results to the SPP website. The latest ELCC curves will be utilized in the allocation portion of the applicable ESR ELCC study as outlined in Attachment AA of the SPP OATT.

7.1.8 ACCREDITATION FOR WIND AND SOLAR UNITS

Effective June 1, 2026, for the SPP East BAA's 2026 Summer Season, Accredited Capacity for wind and solar resources will be established using ELCC methodology and the methodology in section 7.1.8 will become obsolete. The recommended methodology to evaluate the net planning capability established for wind or solar facilities shall be determined on a monthly basis, as stated below. If a member's desire to use a more restrictive methodology to evaluate the net capability of wind or solar they may do so, however net capability determined by the alternative methodology employed cannot credit the wind or solar with a capability greater than determined with the methodology stated below:

- 1) Assemble all available hourly net power output (MWh) data measured at the system interconnection point.
 - (a) Select the hourly net power output values occurring during the top 3% of load hours for the SPP Load Serving Entity (LSE) for each month of each year for the evaluation period.
 - (b) Select the hourly net power output value that can be expected from the facility 60% of the time or greater. For example, for a five-year period with the 110 hourly net power output values ranked from highest to lowest, the capacity of the facility will be the MW value in the 65th data point.
 - (c) A seasonal or annual net capability may be determined by selecting the

appropriate monthly MW values corresponding to the LSE's peak load month of the season of interest (e.g. 22 hours for a typical 30-day month and 110 hours for a five-year period).

- (d) Facilities in commercial operation three years or less:
 - i. The data must include the most recent three years.
 - ii. Values may be calculated from wind or solar data, if measured MW values are not yet available. Wind data correlated with a reference tower beyond 50 miles is subject to Supply Adequacy Working Group approval. Solar data correlated with a reference measuring device beyond two hundred miles is subject to Supply Adequacy Working Group approval. For calculated values, at least one year must be based on site specific data.
 - iii. If the LSE chooses not to perform the net capability calculations as described above during the first 3 years of commercial operation, the LSE may submit 5% for wind facilities and 10% for solar facilities of the site facility's nameplate rating.
- (e) Facilities in commercial operation 4 years and greater:
 - i. The data must include all available data up to the most recent 10 years of commercial operation.
 - ii. Only metered hourly net power output (MWh) data may be used.
 - iii. After three years of commercial operations, if the LSE does not perform or provide the net capability calculations to the Transmission Provider as described above, then the net capability for the resource will be 0 MW.
- (f) The net capability calculation shall be updated at least once every three years.

7.1.9 **ACCREDITATION FOR ENERGY STORAGE RESOURCES**

Effective June 1, 2026, for the SPP East BAA's 2026 Summer Season, Accredited Capacity for Energy Storage Resources will be established using ELCC methodology and the methodology in section 7.1.9 will become obsolete. ESRs with a time duration of four hours or greater will be accredited based on the four hour minimum continuous duration requirement with no adjustments. ESR of less than four hours will have their accreditation adjusted to a four hour equivalent (e.g. A two hour duration ESR will be adjusted to 50% of its rating to meet the four hour minimum continuous duration requirement).

7.1.10 **TESTING REQUIREMENTS AND ACCREDITATION FOR DEMAND RESPONSE PROGRAMS RESOURCES AND RDRPS**

~~Demand Response Programs are as defined in Attachment AA of the SPP OATT.~~

7.1.10.1 **Capability Test**

~~Demand Response Programs will be accredited based on a sustainable level of reduction as outlined in the SPP Business Practices. The amount of load reduction shall be 100% of the claimed capability, or the load shall be reduced to an amount that is less than the load forecast of the Demand Response Programs customer by an amount equivalent to 100% of the claimed capability.~~

~~If a Demand Response Program is deployed during the Summer Season, the deployment of that Demand Response Program can suffice in place of the Capability Test. Demand Response~~

Programs in the first year of operation or expansion of an existing program will be reported at 50% of the expected capability, unless validated by testing the program to 100% of the claimed capability prior to May 15. This test will be similar to a Capability Test but will be conducted outside of the peak season. Testing outside of the peak season will only be considered a Capability Test during the first year of operation or existing program expansion. An Operational Test shall be performed during the upcoming Summer Season and reported to the Transmission Provider in accordance with the SPP Business Practices.

The seasonal capability established for Demand Response Programs shall be adjusted in accordance with the Demand Response Programs Reporting and Documentation Procedures given in the SPP Business Practices. The Capability Test for each Demand Response Resource and RRDRP must perform a SPP-directed test once every three years. If Demand Response Capacity is qualified in accordance with Attachment AA of the SPP OATT for both the Summer Season and Winter Season, a Capability Test must be performed in each season.

For Demand Response Resource and RRDRP, the following test procedures will apply:

1. SPP operations staff will inform the Market Participant or LRE that a test will be conducted with no less than 24-hours' notice.
2. The Market Participant or LRE will then inform SPP operations staff whether the test should proceed.
3. The Market Participant or LRE will be required to update the available capacity of the Demand Response Resource or RRDRP to be tested. Submitting zero available capacity will result in a reschedule of the Capability Test.
4. SPP operations staff will then direct the Market Participant or LRE to perform the Capability Test at any point during the testing period.
5. The amount of load reduced during the testing period shall be the demonstrated Net Generating Capability of the Demand Response Resource or RRDRP.
6. A Market Participant or LRE is not permitted to self-deploy during the Capability Test to ensure the program can reduce load during the allotted time.
7. Market Participants or LREs will be allowed to reschedule the Capability Test one time per year.

If a Demand Response Resource or RRDRP is deployed at the instruction of SPP and such deployment was not originally designated as a Capability Test, then the actual deployment may suffice as a Capability Test if it meets the Capability Test requirements. The entity must inform SPP the actual deployment suffices as a Capability Test and SPP must acknowledge the actual deployment suffices.

The seasonal capability established for a Demand Response Resource or RRDRP shall be adjusted in accordance with the Demand Response Reporting and Documentation Procedures given in the SPP Business Practices. The Demand Response Resource or RRDRP's Net Generating Capability shall not be greater than the highest expected load for the applicable Demand Response Resource or RRDRP. A Capability Test may be conducted outside of the applicable peak season but shall only be considered a Capability Test during the first year of operation. It shall follow the same parameters as the Capability Test requirements as stated above. Every Demand Response

Resource and RRDRP will be required to perform one SPP-directed test during the first peak operating season after initial enrollment.

If a Capability Test cannot be conducted prior to the first applicable season for which the resource is qualified to meet a Load Responsible Entity's Pre-Season Requirement in accordance with Attachment AA of the SPP OATT, the following shall apply in the first year of operation:

1. An entity may claim up to 50% of the expected capability for the Demand Response Resource or RRDRP if design output documentation for the first peak operating season is provided in the resource adequacy Workbook. The design output documentation must contain the expected Net Generating Capability as well as all other reporting and documentation information outlined in the SPP Business Practices.
- 4-2. If an entity does not provide design output documentation or conduct a test prior to the first year of operation, then the Demand Response Capacity of such resources shall be limited at 0% of the expected capability.

7.1.10.2 Operational Test

An Operational Test is used to demonstrate the ability of a program. An Operational Test is a yearly demonstration on the functional ability of a Demand Response Resource or RRDRP. The Operational Test for each Demand Response Resource and RRDRP must perform a SPP-directed test once every year. Operational Test for a Demand Response Program Resource or RRDRP shall be conducted at a minimum of 50-90% of programs the resource's claimed Net Generating eCapability for a minimum of one hour.

If a Demand Response Program Resource or RRDRP is deployed during the Summer Season, the deployment of that Demand Response Program such resource can suffice in place of the Operational Test if it meets 90% of the Net Generating Capability for the applicable season. If a resource has both Summer Season and Winter Season Demand Response Capacity, the Operational Test must be conducted at a minimum of 90% of both the Summer Season and Winter Season Net Generating Capability.

The Operational Test must be completed in a 12-month period beginning on May 15 of each year and can be conducted in any season for the Summer Season. An Operational Test must be completed in a 12-month period beginning on November 15 of each year and can be conducted in any season for the Winter Season. Test results shall be compiled and submitted via the Workbook submittal process as outlined in Attachment AA of the SPP OATT.

7.1.10.3 Frequency of Testing

Capability Tests shall be conducted at least once every five years. Operational Tests shall be conducted once every year during the LRE's peak season. New programs or programs undergoing a physical or operational modification which could impact capability shall be given a Capability Test.

7.1.10.4 — Procedures for Accrediting Demand Response Programs

Capability and Operational Tests shall be completed during a peak load period in the Summer Season.

The load reduction shall be measured and reported in accordance with the SPP Business Practices.

7.1.10.3 Accreditation

The Accredited Capacity for Demand Response Resource and RRDRP shall be established in accordance with Attachment AA of the SPP OATT and SPP Business Practices. The Emergency DR Limit applied in accordance with Attachment AA of the SPP OATT shall be 3,500 MW for the SPP East Balancing Authority Area.

7.1.11 FUEL SUPPLY

Assurance of having desired generating capacity depends, in part, on the availability of an adequate and reliable fuel supply.

Commented [SP9]: RR703 Demand Response

- Approved by the RSC on:
- Approved by the Board on:
- Posted with FERC: 4/17/2026
 - FERC Docket #: ER26-2249-000
- Requested FERC Effective Date: 1/1/2027

7.2 RATING OF TRANSMISSION FACILITIES

All transmission facilities, referenced in Section 5.1, shall have a Normal Rating and an Emergency Rating. Each SPP member shall provide the Normal Rating, Emergency Rating, and associated time durations to the SPP Planning Coordinator upon request.

SPP members may develop dynamic or seasonal ratings based on windspeed, ambient temperature or other variables if permitted by their individual Facility Rating methodology document. The following sections provide guidance and recommendations to Transmission Owners (for voluntary use) in developing their respective Facility Ratings methodology document.

7.2.1 **POWER TRANSFORMER**

Power transformer ratings are discussed in ANSI/IEEE C5791, IEEE Guide for Loading Mineral-Oil-Immersed Power Transformers. Every transformer has a distinct temperature rise capability used in setting its nameplate rating (either 55°C or 65°C). These temperature rise amounts reflect the average winding temperature rise over ambient that a transformer may operate on a continuous basis and still provide normal life expectancy.

7.2.1.1 **Normal Rating**

The recommended Normal Rating for power transformers is its highest nameplate rating. The nameplate rating includes the effects of forced cooling equipment if it is available. For multi-rated transformer (OA/FA, OA/FA/FA, OA/FOA/FOA, OA/FA/FOA) with all or part of forced cooling inoperative, nameplate rating used is based upon the maximum cooling available for operation. Normal life expectancy will occur with a transformer operated at continuous nameplate rating.

7.2.1.2 **Emergency Rating**

When operated for one or more load cycles above nameplate rating, the transformer insulation deteriorates at a faster rate than normal. The recommended Emergency Rating for power transformers is a minimum of 100% of its highest nameplate rating. Member systems may use a higher Emergency Rating if they are willing to experience more transformer loss-of-life.

7.2.1.3 **Loss of Life**

In ANSI/IEEE C57.91, a 65°C rise transformer can operate at 120% for an eight-hour peak load cycle and will experience a 0.25% loss of life. If a 65°C rise transformer experiences four incidents where it operates at or below 120% for an eight-hour peak load cycle, it will still be within the target of 1% loss of life per year. In ANSI/IEEE C57.91, a 55°C rise transformer can operate at 123% for an eight-hour peak load cycle and will experience a 0.25% loss of life. Likewise, if a 55°C rise transformer experiences four incidents where it operates at or below 123% for an eight-hour peak load cycle, it will still be within the target of 1% loss of life per year.

7.2.1.4 Ambient Temperature

Average ambient temperature is an important factor in determining the load capability of a transformer since the temperature rise for any load must be added to the ambient to determine operating temperature. Transformers designed according to ANSI standards use a 30°C average ambient temperature (average temperature for 24 consecutive hours) when setting nameplate rating. Transformer overloads can be increased at lower average ambient temperatures and still experience the same loss of life. This allows seasonal ratings with higher Normal and Emergency Ratings. However, seasonal transformer ratings are optional. In ANSI/IEEE C57.91, transformers can be loaded above 110% and experience no loss of life when the average ambient temperature is below 78°F. By not having seasonal ratings, the four occurrences that contribute to loss of life are limited to days when the average ambient temperature exceeds 78°F. The Power Transformer Rating Factors include:

- 1) Nameplate rating, normal loss of life for 55°C and 65°C rise transformers with cooling equipment operating.
- 2) Average ambient temperature, 30°C.
- 3) Equivalent load before peak load, 90% of nameplate rating.
- 4) Hours of peak load, eight hour load cycle.
- 5) Acceptable annual loss of life, 1%.

7.2.2 OVERHEAD CONDUCTOR

Overhead conductor ratings are discussed in IEEE Standard 738, IEEE Standard for Calculating the Current-Temperature Relationship of Bare Overhead Conductors. Ampacity values are to be determined using the fundamental heat balance equation outlined in the House and Tuttle method. Because of the amount and complexity of the equations, this method lends itself to computer application. The recommended computer programs to be used for this calculation either include the BASIC program listed in Annex B of IEEE Standard 738 or an equivalent program, such as the DYNMAP program which is part of the EPRI TL Workstation™ software package. While tables and graphs may be convenient to use, they fail to take into account the geographic location of the line and often lack either the desired ambient temperature and/or the desired conductor temperature. The use of tables and graphs is not recommended.

7.2.2.1 Conductor Properties

Some computer programs used to compute ampacity values have a conductor property library whereby a user simply specifies the conductor code name and the program will search the conductor property file and select the proper input properties. Those using the BASIC program from Annex B of IEEE Standard 738 or another computer program that does not have a conductor property library should obtain conductor properties from an appropriate data source (Aluminum Electrical Conductor Handbook, EPRI Transmission Line Reference Book 345 kV and Above, Westinghouse Transmission and Distribution Book, etc.).

7.2.2.2 Line Geographic Location

These factors specify the location of the line, its predominant direction and its predominant inclination. These numbers can either be line specific or they can represent a general line within the area.

7.2.2.3 Radiation Properties

The two radiative properties of conductor material are solar absorptivity and infrared emissivity.

Solar Absorptivity The fraction of incident solar radiant energy that is absorbed by the conductor surface. This value should be between 0 and 1. Recommended values are given in the following tables:

Copper Conductors	
Oxidation Level	Absorptivity
None	0.23
Light	0.5
Normal	0.7
Heavy	1.0

Aluminum Conductors	
Service Years	Absorptivity
0<5	.43

Source: Glenn A. Davidson, Thomas E. Donoho, George Hakun III, P. W. Hofmann, T. E. Bethke, Pierre R. H. Landrieu and Robert T. McElhaney, "Thermal Ratings for Bare Overhead Conductors," IEEE Trans., PAS Vol. 88, No.3, pp. 200-05, March 1969.

Infrared Emissivity The ratio of infrared radiant energy emitted by the conductor surface to the infrared radiant energy emitted by a blackbody at the same temperature. This value should be between 0 and 1. Recommended values are given in tables below:

Copper Conductors	
Oxidation Level	Emissivity
None	0.03
Light	0.3
Normal	0.5
Heavy	0.8

Aluminum Conductors	
Service Years	Emissivity
0	0.23
5-10	0.82
10-20	0.88
	0.90

Source: W. S. Rigdon, H. E. House, R. J. Grosh and W. B. Cottingham, "Emissivity of Weathered Conductors After Service in Rural and Industrial Environments," AIEE Trans., Vol. 82, pp. 891-896, Feb. 1963.

7.2.2.4 Weather Conditions

Ambient temperature represents the maximum seasonal temperature the line may experience for summer and winter conditions. Section 10.1 in this SPP Planning Criteria contains a methodology to compute maximum ambient temperature. Wind speed is assumed at two feet per second (1.4 mph) or higher. Wind direction is assumed perpendicular to the conductor.

7.2.2.5 Maximum Conductor Temperature

The selection of a maximum conductor temperature affects both the operation and design of transmission lines. Existing transmission lines were designed to meet some operating standard that was in effect at the time the line was built. That standard specified the maximum conductor temperature which maintained acceptable ground clearance while allowing for acceptable loss of strength. Over time, the required amount of ground clearance and the maximum conductor temperature needed to maintain acceptable ground clearance have changed. The changes are reflected in the revisions that have been made to the National Electric Safety Code (NESC) over the years. Although this Criteria recommends a maximum conductor temperature that could be met by current line design practices, consideration should be given to existing lines that were built according to an earlier standard. This rating criteria recommends a maximum conductor temperature (for both normal and emergency operating conditions) that should be used if developing seasonal ratings. For those existing lines that were designed to meet an earlier

standard, it is the responsibility of the line owner to establish a rating that is consistent with the NESC design standards being practiced at the time the line was built. This Criteria specifies the use of maximum conductor temperatures that either maintain acceptable ground clearance requirements from earlier NESC's or meet the temperature recommendations in SPP Planning Criteria section 7.2.2.6, whichever is lower.

7.2.2.6 Determination of Maximum Conductor Temperature

The maximum conductor temperature for Normal Ratings may be limited by conductor clearance concerns. Normal Ratings are at a level where loss of strength is not a concern. The maximum conductor temperature for Emergency Ratings has both conductor clearance and loss of strength concerns. By setting a maximum conductor temperature and the length of time a conductor may operate at this temperature, the maximum allowable loss of strength over the life of the conductor is prescribed. Unless conductor clearance concerns dictate otherwise, at least the following maximum conductor temperatures can be used. This allows for the efficient utilization of the transmission system while accepting minimal risk of loss of conductor strength during emergency operating conditions. These conductor temperatures are a result of the examination of SPP members' practices.

	Maximum Conductor Temperature	
	Normal Rating	Emergency Rating
ACSR	85°C	100°C
ACAR	85°C	100°C
Copper	85°C	100°C
Copperweld	85°C	100°C
AAC	85°C	100°C
AAAC	85°C	100°C
SSAC	200°C	200°C
Note: Annealing of copper and aluminum begins near 100°C		

7.2.2.7 Hours of Operation at Emergency Rating

The effect of conductor heating due to operating at the maximum temperature during emergency conditions is cumulative. If a conductor is heated under emergency loading for four hours, eight times during the year, the total effect is nearly the same as heating the conductor continuously at the temperature for 32 hours. Using a useful conductor life of 30 years, the conductor will have been heated to the maximum temperature for 1,000 hours. For an all-aluminum conductor (AAC), this results in a 7% reduction from initial strength. Since the steel core of an ACSR conductor is essentially unaffected by the temperature range considered for emergency loadings, for an ACSR

conductor, this results in a 3% reduction from initial strength. Both of these amounts are acceptable loss of strength. The daily load cycle for operating at the Emergency Rating should not exceed 4 hours. This load cycle duration for conductors operating at the Emergency Rating is more restrictive than power transformers because power transformers have a delay in the time required to reach a stable temperature following any change in load (caused by a thermal lag in oil rise) and because seasonal ratings should allow transmission lines to achieve a maximum conductor temperature throughout the year, not just days when the ambient exceeds 78°F.

7.2.3 UNDERGROUND CABLES

Ampacities are calculated by solving the thermal equivalent of Ohm's Law. Conceptually, the solution is simple, however the careful selection of the values of the components of the Facility is necessary to ensure an accurate ampacity calculation. The recognized standard for almost all steady-state ampacity calculations, in the United States, is taken from a publication, "The Calculation of the Temperature Rise and Load Capability of Cable Systems," by J.H. Neher and M.H. McGrath, 1957, hereafter referred to as the Neher-McGrath method. The procedure is relatively simple to follow and has been verified through testing. In recent years, some of the parameters have been updated, but the method is still the basis of all ampacity calculations.

7.2.3.1 Cable Ampacity

Cable ampacity is dependent upon the allowable conductor temperature for the particular insulation being used. Conductor temperature is influenced by the following factors:

- 1) Peak current and load-cycle shape;
- 2) Conductor size, material and construction;
- 3) Dielectric loss in the insulation;
- 4) Current-dependent losses in conductor, shields, sheath and pipe;
- 5) Thermal resistances of insulation, sheaths and coverings, filling medium, pipe or duct and covering, and earth;
- 6) Thermal capacitances of these components of the thermal Facility;
- 7) Mutual-heating effects of other cables and other heat sources; and
- 8) Ambient earth temperatures.

Both steady-state and emergency ampacities depend upon these factors, although Emergency Ratings have a greater dependency upon the thermal capacitances of each of the thermal Facility components.

7.2.3.2 Conductor Temperature

The maximum allowable conductor temperature is 85°C for high-pressure fluid-filled (HPFF), pipe-type cables and 90°C for cross-linked, extruded-dielectric cables.

The table below summarizes allowable conductor temperatures for different insulation materials. Two values are given for each cable insulation. The higher temperature may be used if the thermal environment of the cable is well-known along the entire route, or if controlled backfill is used, or if fluid circulation is present in an HPFF Facility. The maximum conductor

temperatures allowed under steady-state conditions are limited by the thermal aging characteristics of the insulation structure of the cable. For emergency-overload operating conditions, maximum conductor temperatures are also limited by the thermal aging characteristics. The temperature is also limited by the melting temperature range of the insulation structure of the cable, its deformation characteristic with temperatures, restraints imposed by the metallic shield, deformation characteristic of the jacket, and the decrease in ac and impulse strengths with increases in temperature.

Insulation Material	Maximum Temperature	
	Normal	Emergency
Impregnated paper (AEIC CS2-90 for HPFF and HPGF (AEIC CS4-79 for SCLF)	85°C (75°C)	105°C for 100 hr. 100°C for 300 hr.
Laminated paper-polypropylene (AEIC CS2-90)	85°C (75°C)	105°C for 100 hr. 100°C for 300 hr.
Cross-linked polyethylene (AEIC CS7-87)	90°C (80°C)	105°C Cumulative for 1500 hr.
Ethylene-propylene rubber (AEIC CS6-87)	90°C (80°C)	105°C Cumulative for 1500 hr.
Electronegative gas/spacer	Consult manufacturer for specific designs	
*Emergency operation at conductor temperatures up to 130°C may be used if mutually agreed between purchaser and manufacturer and verified by qualification and prequalification tests		

7.2.3.3 Ambient Temperature

The ambient temperature is measured at the specified burial depth for buried cables and the ambient air temperature is used for cables installed above ground. IEC Standard 287-1982 (2-5) recommends that in the absence of national or local temperature data the following should be used:

Climate	Ambient Air Temperature °C	Ambient Ground Temperature °C
Tropical	55	40
Sub-tropical	40	30
Temperate	25	20

The electrical resistance is composed of conductor dc resistance, ac increments due to skin and proximity effects, losses due to induced currents in the cable shield and sheath and induced magnetic losses in the steel pipe. Heat generated in the cable system will flow to ambient earth and then to the earth surface. This heat passes through the thermal resistances of the cable insulation, cable jacket, duct or pipe space, pipe covering and soil. Adjacent heat sources, such as other cables or steam mains, will provide impedance to the heat flow and thus reduce cable ampacity. Further information concerning the components of the ampacity calculations are summarized in Section 10.2 in this SPP Planning Criteria and fully detailed in the EPRI Underground Transmission Systems Reference Book. An example calculation, from the EPRI book, is also provided in this SPP Planning Criteria section 10.2.

7.2.4 SWITCHES

Section 10.4 of this SPP Planning Criteria contains a discussion on developing ratings for switches. In general, switches have seasonal ratings that are a function of the maximum ambient temperature. A switch part class designation is used to differentiate loadability curves that give factors which can be multiplied by the rated continuous current of the switch to determine temperature adjusted Normal and four hour Emergency Ratings. The summer Normal and Emergency Ratings for switches can be computed by selecting the appropriate loadability factor curve for the switch part class, reading the loadability factors that are appropriate for the summer maximum ambient temperature (40°C or the summer maximum ambient temperature determined in section 10.1 in this SPP Planning Criteria), and multiplying the continuous current ratings by the loadability factor. The switch winter normal and emergency ratings can be computed by multiplying the continuous current rating by the normal and emergency loadability factors that are appropriate for the winter maximum ambient temperature (0°C or the winter maximum ambient temperature determined in section 10.1 in this SPP Planning Criteria). Section 10.4 of this SPP Planning Criteria contains loadability factor curves (both normal and emergency) for various switch part classes. The ANSI/IEEE standard referenced in Section 10.4 of this SPP Planning Criteria allows for Emergency Ratings to be greater than Normal Ratings. The Emergency Rating does not need to be greater than the Normal Rating.

7.2.5 WAVE WRAPS

Section 10.4 of this SPP Planning Criteria contains a discussion on developing ratings for wave traps. The two types of wave traps are the older air-core type and the newer epoxy-encapsulated type. In general, both types have a continuous current rating based on a 40°C maximum ambient temperature. Both types have a loadability factor that can be used to determine seasonal ratings that are a function of the maximum ambient temperature. However, the older air-core type has another loadability factor that can be used to determine a four hour Emergency Rating that is also a function of the maximum ambient temperature. The newer epoxy encapsulated type does not have an Emergency Rating.

7.2.6 CURRENT TRANSFORMERS

Section 10.5 in this SPP Planning Criteria contains a discussion on developing ratings for current transformers. The two types of current transformers are the separately-mounted type and the bushing type. In general, both types have a continuous current rating based on a 30°C average ambient temperature.

7.2.6.1 Separately Mounted Current Transformers

The separately-mounted type has an ambient-adjusted continuous thermal current rating factor that can be multiplied by the rated primary current of the current transformer to determine seasonal ratings. Separately-mounted current transformers do not have Emergency Ratings.

7.2.6.2 Bushing Current Transformers

Bushing current transformers are subject to and influenced by the environment of the power apparatus in which they are mounted. Bushing current transformers can be located within circuit breakers and power transformers. Since bushing current transformers are subject to the environment within the power apparatus, they do not have ambient adjusted continuous thermal current rating factors. Rather, if the primary current rating of the ratio being used is less than the continuous current rating of the breaker or the power transformer, this restricts the breaker or power transformer to operate below its rated current which reduces the current transformer temperature. This allows the current transformer to be operated at a continuous thermal rating factor greater than 1.0. Having a bushing current transformer whose primary current rating of the ratio being used is less than the continuous current rating of the breaker or the power transformer is an unusual case. However, the formula to develop the rating factor for this case is located in Section 10.5 in this SPP Planning Criteria. Although bushing current transformers have some short-term emergency overload capability, it must be coordinated with the overall application limitation of the other equipment affected by the current transformer loading. Consequently, this criteria does not recognize an Emergency Rating for bushing current transformers.

7.2.7 CIRCUIT BREAKERS

Section 10.6 in this SPP Planning Criteria contains a discussion on developing ratings for circuit breakers. This discussion centers on the use of specific circuit breaker design information to set seasonal and Emergency Ratings. This design information is not readily available to the owners of such equipment. To use the rating methodology discussed in Section 10.6 in this SPP Planning Criteria would require contacting the manufacturer for detailed design information for each circuit breaker being rated. Rather than doing that, this rating criteria specifies that the nameplate rating should be used for seasonal Normal and Emergency Ratings. The nameplate rating is based on a maximum ambient temperature of 40°C. If a circuit breaker is found to be a limiting element and is experiencing loadings that limit operations, a member system may pursue the methodology outlined in Section 10.6 in this SPP Planning Criteria to determine the circuit breakers seasonal Normal and Emergency Rating.

7.2.8 RATINGS OF SERIES AND REACTIVE ELEMENTS

The series transmission elements rating will be in amps, ohms, and MVA. The series transmission elements current (amps) rating will be taken as the minimum rating of all internal components (e.g. breakers) that are in series with the interconnected transmission Facility. Shunt reactive elements (e.g. capacitors, reactors) MVA ratings will be based on the nominal transmission interconnecting voltage.

7.2.9 RATINGS OF ENERGY STORAGE DEVICES

The available real power rating, reactive power rating, control points, and availability of each electrical energy storage device will be provided to SPP upon request.

7.2.10 FACILITY RATING ISSUES

7.2.10.1 *Dynamic (Real Time) Ratings*

The calculation of static thermal ratings specified in SPP Planning Criteria section 7.2.2.6 uses worst case thermal and operational factors and therefore apply under all conditions. Often times, these worst case thermal and operational factors do not all occur at the same time. Consequently, a static rating may understate the thermal capacity of the Facility. For operation purposes, some members have elected to monitor the factors that affect Facility Ratings and use this information to set dynamic ratings. A member can develop and use a rating that exceeds the static thermal rating for operating purposes. The ratings developed by using this criteria are not intended to restrict daily operations but set a minimum rating that can be increased when factors for determining the equipment rating have changed. However, if Facility Ratings are changed dynamically, the required clearances should still be met.

7.2.10.2 *Non Thermal Limitations*

There may be instances when the flow on a transmission is limited by factors other than the thermal capacity of its elements. The limit may be caused by other factors such as dynamics, phase angle difference, relay settings or voltage limited.

7.2.10.3 *Tie Lines*

When a tie line exists between two member systems, use of this criteria should result in a uniform Facility Rating that is determined on a consistent basis between the two systems. For tie lines between a SPP member and a non-member, the member should follow this criteria to rate the elements owned by them and should coordinate the rating of the tie line with the non-member system such that it utilizes the lowest rating between the two systems.

7.2.10.4 *Ratings Inconsistencies*

A member may have a contractual interest in a joint ownership transmission line whereby the capacity of the line is allocated among the owners. The allocated capacity may be based upon the thermal capacity of the line or other considerations. Members should use good faith effort to amend their transmission line agreements to reflect the effects of new Facility Ratings. There may exist other transmission agreements or regulatory mandates that use the thermal capacity of transmission Facilities in allocation of cost and determination of network usage formulas (for example, the MW-mile in ERCOT). These agreements and mandates may specify a methodology and/or factors for computing thermal capacity used in the formulas. Since these amounts are only used in assignment of cost or usage responsibility and not in actual operations of the transmission system, there is no conflict with using a different set of ratings for this specific purpose.

7.2.10.5 Damaged Equipment

There may be instances when a de-rating of a transmission line element is required due to damaged equipment. The limit may be caused by such factors as broken strands, damaged connectors, failed cooling fans, or other damage reducing the thermal capability.

7.2.11 HVDC

The ratings of each HVDC facility will be provided to SPP upon request. HVDC facilities are comprised of, but not limited to, converters, transformers, conductors, and associated equipment such as filters, switches, and buses. The ratings may be based on the manufacturer's specifications.

8. SYSTEM PROTECTION REQUIREMENT

All transmission facilities shall have appropriate system protection and control equipment and settings.

8.1 DEFINITIONS

8.1.1 EFFECTIVE DATE

A Remedial Action Scheme's (RAS) Effective Date shall be determined by the RAS Sponsor submitting a RAS proposal for approval to indicate the date the RAS shall be placed into service and will be utilized for the screening process and subsequent in-service tracking by SPP staff.

8.1.2 EQUIVALENT ACHIEVEMENT OF SYSTEM PERFORMANCE REQUIREMENT

An Equivalent Achievement of System Performance Requirement shall mean when comparing the mitigation yielded by implementation of a RAS to that of a permanent transmission system change or upgrade, the transmission system shall maintain the equivalent system performance requirements of the SPP Planning Criteria and NERC TPL-001, Table 1.⁴

8.1.3 EXTENDED USE RAS

An Extended Use RAS is a long-term mitigation that can be utilized in lieu of a transmission change or upgrade. An Extended Use RAS may serve as a Corrective Action Plan (CAP) that prevents a failure to meet system performance requirements specified by applicable NERC Reliability Standards, SPP Planning Criteria, and/or other applicable SPP Business Practices.

8.1.4 TEMPORARY RAS

A Temporary RAS is an interim mitigation that can be utilized when a transmission change or upgrade has been identified. A Temporary RAS may serve as a CAP that prevents a failure to meet

⁴ <https://www.nerc.com/pa/Stand/Pages/ReliabilityStandards.aspx>

system performance requirements specified by applicable NERC Reliability Standards, SPP Planning Criteria, and/or other applicable SPP Business Practices.

8.1.5 TEMPORARY RAS SERVICE DURATION

The Temporary RAS Service Duration is the period of service during which operation of the Temporary RAS is permitted. The Temporary RAS Service Duration shall not be less than 12 months and shall not exceed the period between the RAS Effective Date and the expected in-service date of the permanent transmission system change or upgrade directed by the SPP Transmission Provider. No Temporary RAS Service Duration shall exceed 30 months, unless approved for adjustment (section 8.3.2.5). The Temporary RAS is placed into RAS Retirement (section 8.1.7) upon conclusion of the Temporary RAS Service Duration or completion of the identified transmission change or upgrade.

8.1.6 RAS SPONSOR

The RAS Sponsor is a party that requests approval for the use of a RAS on the SPP transmission system. The RAS Sponsor shall be a RAS Entity (as defined in PRC-012 as Transmission Owner, Generator Owner, or Distribution Provider that owns all or part of a RAS) or have a written agreement with the RAS Entity(ies) of the sponsored RAS to seek its approval.

8.1.7 RAS RETIREMENT

The RAS Retirement is a permanent deactivation and decommissioning of the RAS.

8.2 UNDER-FREQUENCY LOAD SHEDDING

NERC Reliability Standard PRC-006 requires each Planning Coordinator to develop an under-frequency load shedding (UFLS) program for its area. The SPP UFLS Plan is designed to meet the NERC UFLS requirements for the SPP Planning Coordinator area. Each UFLS Entity in the SPP Planning Coordinator area, shall follow the criteria and requirements defined in the SPP UFLS Plan document, which can be found on the SPP website.

8.3 REMEDIAL ACTION SCHEME APPROVAL

All RAS in the SPP Reliability Coordinator and Planning Coordinator areas are subject to PRC-012 and its associated requirements. Prior to placing a new RAS into service, completing a five-year RAS evaluation, functionally modifying a RAS, retiring a RAS, or adjusting a Temporary RAS Service Duration, each RAS shall meet the RAS screening criteria (section 8.3.1), and complete the applicable approval process (section 8.3.2).

8.3.1 RAS SCREENING CRITERIA

All RAS shall meet the following applicable screening criteria prior to consideration for approval:

8.3.1.1 *Screening Criteria for Transmission Owner and Transmission Operator Requirements*

A RAS shall not contradict Transmission Owner interconnection requirements, Zonal Planning Criteria (ZPC), nor Transmission Operator procedures. The requesting RAS Sponsor shall coordinate all NERC Reliability Standard compliance responsibilities (*e.g.*, FAC-001, PRC-012,

etc.), and related compensation for compliance obligations associated with the implementation of the RAS with applicable Transmission Owners and Transmission Operators.

8.3.1.2 Screening Criteria for an Extended Use RAS

An Extended Use RAS may be utilized in lieu of a transmission change or upgrade and considered for approval if each of the follow criteria are met:

- 1) An upgrade is not required through any planning process;
- 2) Other mitigation plans have been considered;
- 3) The RAS is demonstrated to meet the Equivalent Achievement of System Performance Requirements; and,
- 4) Extended Use RAS actions shall not instantaneously remove load or generation exceeding the thresholds defined in Section 15, Appendix 1 - SPP RAS Thresholds.

If a new Extended Use RAS does not pass the screening criteria it will not be considered for approval. If an existing Extended Use RAS does not pass the screening criteria SPP will coordinate with the RAS Sponsor and Transmission Owner(s)/Operator(s) to reliably retire the RAS.

All existing RAS with grandfathered approval from the Midwest Reliability Organization (MRO) and recorded in the SPP RAS database as of the PRC-012 Implementation Plan Effective Date (01 Jan 2021) shall be grandfathered as an Extended Use RAS and not subject to the above screening criteria.

8.3.1.3 Screening Criteria for a Temporary RAS

A Temporary RAS is a RAS that may be utilized on a temporary basis for an approved Temporary RAS Service Duration no longer than necessary to allow required permanent transmission system changes or upgrades to be placed into service, and may be considered for approval if each of the following criteria are met:

- 1) The RAS Sponsor specifies the permanent transmission system change or upgrade for which the RAS is a comparable interim solution;
- 2) The RAS is demonstrated to meet Equivalent Achievement of System Performance Requirements as the specified permanent transmission system change or upgrade;
- 3) The RAS has an appropriate Temporary RAS Service Duration; and,
- 4) Temporary RAS actions shall not instantaneously remove load or generation exceeding the thresholds defined in Appendix 1.

8.3.2 RAS APPROVAL PROCESS

During the lifecycle of each RAS proposed or implemented within the SPP transmission system, the following approval processes shall be used:

8.3.2.1 New RAS Approval Process

All new RAS, whether Extended Use or Temporary, shall perform the following approval process:

- 1) The RAS Sponsor shall submit all materials required by the applicable SPP PRC-012 process to SPP.
- 2) SPP shall screen the new RAS for approval in accordance with applicable affected sections of 8.3.1 and present RAS materials to the applicable working groups. If the RAS does not pass the screening SPP shall inform the RAS Sponsor and it will not be considered for approval.
- 3) The new RAS shall be reviewed by each of the:
 - (a) System Protection and Control Advisory Committee (SPCAG);
 - (b) Transmission Working Group (TWG); and,
 - (c) Operating Reliability Working Group (ORWG).
- 4) Each working group review shall consist of a technical examination including, but not limited to the:
 - (a) RAS design and planned operational details;
 - (b) RAS intended purpose and proposed mitigation actions;
 - (c) Suitability of the RAS to meet Equivalent Achievement of System Performance Requirements.
- 5) Upon review by the working groups, each working group shall make a recommendation for new RAS approval or denial to the Markets and Operations Policy Committee (MOPC).
- 6) The MOPC shall approve or deny all new RAS.

8.3.2.2 Functional Modification to Existing RAS Approval Process

Any functional modifications to an existing RAS, whether Extended Use or Temporary, shall perform the following approval process:

- 1) The RAS Sponsor shall submit all materials required by the applicable SPP PRC-012 process to SPP.
- 2) SPP shall screen the RAS functional modification for approval in accordance with applicable affected sections of 8.3.1 and present RAS materials to the applicable working groups. If the RAS does not pass the screening SPP shall inform the RAS Sponsor and it will not be considered for approval.
- 3) The RAS functional modification shall be reviewed by each of:
 - (a) SPCAG;
 - (b) TWG; and,
 - (c) ORWG.
- 4) Each working group review shall consist of a technical examination including, but not limited to the:
 - (a) Functionally modified RAS design and planned operational details;
 - (b) Intended purpose of the RAS, proposed mitigation actions, and effects of functional modification to both;
 - (c) Suitability of the functionally modified RAS to meet Equivalent Achievement of System Performance Requirements.
- 5) Upon review by the working groups, each working group shall make a recommendation to approve or deny the RAS functional modification to the MOPC.
- 6) The MOPC shall approve or deny all functional modifications to existing RAS.

8.3.2.3 Existing RAS Review Process

Each Extended Use RAS shall be reviewed according to the following process every five years consistent with PRC-012. The purpose of the five-year evaluation conducted by the SPP Planning Coordinator is to ensure the Extended Use RAS continues to mitigate the system conditions or contingencies for which it was designed, avoids adverse interactions with other protection and control systems, and satisfies system performance requirements for inadvertent operation and single component failure.

- 1) The RAS Sponsor shall submit updated materials required by the applicable SPP PRC-012 process to SPP.
- 2) SPP shall screen the RAS for review in accordance with applicable affected sections of 8.3.1. If the RAS does not pass the screening SPP shall inform the RAS Sponsor and it will not be considered for approval.
- 3) SPP Planning Coordinator shall conduct the five-year RAS evaluation consistent with PRC-012 Requirement R4. Upon completion of its review, the SPP Planning Coordinator shall present five-year RAS evaluation materials to the applicable working groups, including any CAP(s) identified to correct deficiencies during its five-year RAS evaluation.
- 4) The five-year RAS evaluation materials shall be reviewed by each of:
 - (a) SPCAG;
 - (b) TWG; and,
 - (c) ORWG.
- 5) Each working group review shall consist of a technical examination consistent with PRC-012 Requirement R4 including, but not limited to the:
 - (a) RAS design and operational details;
 - (b) RAS intended purpose;
 - (c) Continued suitability of the RAS to meet Equivalent Achievement of System Performance Requirements;
 - (d) Deficiencies and CAP(s) identified during the five-year RAS evaluation.
- 6) Each working group shall make a recommendation to approve or reject the SPP PC five-year RAS evaluation.
- 7) The SPP PC shall provide the results of its evaluation to the SPP Reliability Coordinator and recommend CAP(s), if applicable.
- 8) SPP RC shall approve or deny any required CAP(s) identified during a five-year RAS evaluation.

8.3.2.4 Retirement of RAS Approval Process

Prior to RAS Retirement of an Extended Use RAS or a Temporary RAS before completion of its Temporary RAS Service Duration, the following approval process shall be performed. A Temporary RAS shall not require approval for RAS Retirement at the conclusion of its Temporary RAS Service Duration and shall be removed from service.

- 1) The RAS Sponsor shall submit all materials required by the applicable SPP PRC-012 process to SPP.

- 2) SPP shall screen the RAS Retirement for approval in accordance with applicable sections of the SPP PRC-012 process and present RAS materials to the applicable working groups. If the RAS does not pass the screening SPP shall inform the RAS Sponsor and it will not be considered for approval.
- 3) The RAS Retirement shall be reviewed by each of:
 - (a) SPCAG;
 - (b) TWG; and,
 - (c) ORWG.
- 4) Each working group review shall consist of a technical examination including, but not limited to the:
 - (a) Suitability of the RAS retirement; and,
 - (b) Expected achievement of required transmission system performance.
- 5) Upon review by the working groups, each working group shall make recommendation to approve or deny the RAS Retirement to the MOPC.
- 6) The MOPC shall approve or deny all RAS Retirements of an Extended Use RAS or a Temporary RAS before completion of its Temporary RAS Service Duration.

8.3.2.5 Adjustment of Temporary RAS Service Duration Approval Process

Delays placing permanent transmission system changes or upgrades into service may necessitate modifications to a Temporary RAS Service Duration. The RAS Sponsor is responsible for determining the need for an adjustment to the Temporary RAS Service Duration. The SPP Reliability Coordinator may approve a one-time, not to exceed six month extension to a Temporary RAS Service Duration without conducting the following approval process and shall inform all applicable working groups of the extension. All proposed adjustments to a Temporary RAS Service Duration exceeding six months shall perform the following approval process. No single Temporary RAS Service Duration adjustment shall exceed 18 months or the period expected for the permanent transmission system change or upgrade to be placed into service, whichever is less. No more than two adjustments of any length shall be approved for a Temporary RAS Service Duration. No two combined adjustments shall exceed a total of 24 months.

- 1) The RAS Sponsor shall submit updated materials required by the applicable SPP PRC-012 process to SPP.
- 2) SPP Planning Coordinator shall determine whether the adjustment to the Temporary RAS Service Duration will cause the Temporary RAS to meet the requirements to conduct a five-year RAS evaluation consistent with PRC-012 Requirement R4.
- 3) SPP shall screen the RAS for review in accordance with applicable affected sections of 8.3.1 and present RAS materials supporting the Temporary RAS Service Duration adjustment to the applicable working groups. If the RAS does not pass the screening SPP shall inform the RAS Sponsor and it will not be considered for approval.
- 4) The RAS materials supporting the Temporary RAS Service Duration adjustment shall be reviewed by each of:
 - (a) SPCAG;
 - (b) TWG; and,
 - (c) ORWG.

- 5) Each working group review shall consist of a technical examination consistent with PRC-012 Requirement R4 including, but not limited to the:
 - (a) RAS design and operational details;
 - (b) RAS intended purpose;
 - (c) Circumstances and impact of delays in placing permanent transmission system changes or upgrades into service;
 - (d) Continued suitability of the RAS to meet Equivalent Achievement of System Performance Requirements.
- 6) Upon review by the working groups, each working group shall make recommendation to approve or deny the Temporary RAS Service Duration adjustment to the MOPC.
- 7) The MOPC shall approve or deny all Temporary RAS Service Duration adjustments exceeding six months.
- 8) If the Temporary RAS Service Duration adjustment is approved by MOPC and the SPP Planning Coordinator determined the Temporary RAS is subject to a five-year evaluation, the Existing RAS Review Process (section 8.3.2.3) shall be used to approve the five-year evaluation of the Temporary RAS.

Should an active temporary RAS not request an extension or a request be denied, SPP will coordinate with the RAS Sponsor and Transmission Owner(s)/Operator(s) to reliably retire the RAS.

8.4 UNDER VOLTAGE LOAD SHEDDING

While there are local under-voltage load shedding (UVLS) schemes implemented by individual UVLS entities, as of the publication date of this criteria, there is no regional UVLS program implemented within the SPP Planning Coordinator area. SPP, as the Planning Coordinator, annually surveys the members to determine if there are any TP UVLS programs or schemes within its Planning Coordinator footprint. If a member plans to implement a UVLS scheme within the SPP footprint, it must submit its information to SPP during this annual survey in order for SPP to determine if a region-wide UVLS program needs to be developed.

9. UNIFORM RATING OF GENERATING EQUIPMENT

Given: Plant site is $39^{\circ}, 15'$ latitude; $94^{\circ}, 58'$ longitude.

Elevation is 750 feet.

Find: Generator rating conditions.

- 1) Determine the three nearest weather stations and temperatures from the ASHRAE Handbook.

Station	Latitude	Longitude	Elevation	Td	Tw
Kansas City, MO	$39^{\circ}, 07'$	$94^{\circ}, 35'$	791	99	75
Atchison, KS	$39^{\circ}, 34'$	$95^{\circ}, 07'$	945	96	77
Topeka, KS	$39^{\circ}, 04'$	$95^{\circ}, 38'$	877	99	75

- 2) Correct the temperatures to the plant elevation:

Station	Elevation above Plant	Effect on Td	Effect on Tw	Corrected Td	Corrected Tw
Kansas City, MO	41	0.2	0.1	99.2	75.1
Atchison, KS	195	1	0.4	97	77.4
Topeka, KS	127	0.6	0.3	99.6	75.3

- 3) Using $39^{\circ}, 00'$; $94^{\circ}, 00'$ as the origin determine the relative location of each point A, B, and C from the plant P. To obtain the east-west dimensions, multiply the longitude minutes from $94^{\circ}, 00'$ by the cosine of plant latitude; $\cos(39.25) = 0.7744$. Doing so, the following coordinates are obtained:

	North	West	Point
Plant	15	44.9	P
Kansas City, MO	7	27.1	A
Atchison, KS	34	51.9	B
Topeka, KS	4	75.9	C

- 4) Calculate the distance between P and points A, B, and C, and then temperatures at P.

$$D_{PA} = ((y_P - y_A)^2 + (x_P - x_A)^2)^{1/2} = 19.5$$

$$D_{PB} = ((y_P - y_B)^2 + (x_P - x_B)^2)^{1/2} = 20.2$$

$$D_{PC} = ((y_P - y_C)^2 + (x_P - x_C)^2)^{1/2} = 32.9$$

So,

$$Ti = \frac{\sum \frac{Ti}{Di}}{\sum \frac{1}{Di}}$$

$$T_d = (T_{dA}/D_{PA} + T_{dB}/D_{PB} + T_{dC}/D_{PC}) / (1/D_{PA} + 1/D_{PB} + 1/D_{PC}) = 98.46^\circ\text{F}$$

$$T_w = (T_{wA}/D_{PA} + T_{wB}/D_{PB} + T_{wC}/D_{PC}) / (1/D_{PA} + 1/D_{PB} + 1/D_{PC}) = 76.01^\circ\text{F}$$

5) The standard barometric pressure for the site is:

$$P_{SB} = 29.921 \times (1 - 6.8753 \times 0.000001 \times \text{Elevation})^{5.2559} = 29.12 \text{ inches Hg}$$

10. TRANSMISSION RATING CRITERIA

10.1 AMBIENT TEMPERATURE

10.1.1 PURPOSE

The purpose of this section is to describe the methodology to be used when determining ambient temperature for circuit rating purposes. This methodology allows for the flexibility of computing ambient temperatures that are location specific and allows for the ease of using default values.

10.1.2 TYPES OF AMBIENT TEMPERATURE

Maximum ambient temperatures are used when calculating seasonal ratings for overhead conductors, disconnect switches, circuit breakers and wave traps. Average ambient temperatures are used when calculating seasonal ratings for power transformers and current transformers.

10.1.2.1 *Maximum Ambient Temperature*

Circuit rating criteria specify the maximum temperature an overhead conductor, switch, circuit breaker, and wave trap may experience. As the ambient temperature increases, the temperature rise that produces the same maximum equipment temperature is reduced.

The reduced temperature rise results in a reduced load carrying capability. Consequently, increases in the maximum ambient temperature does not alter the maximum temperature the equipment was designed to withstand, but it does reduce the current carrying capability for the same maximum equipment temperature.

Selection of the maximum ambient temperature used in rating transmission circuits involves trade-offs. If you select an ambient temperature that is the highest temperature ever recorded for your control area, you may be limiting the use of the transmission system on an ongoing basis for a temperature that has a very small likelihood of occurring. On the other hand, if you select an ambient temperature that is frequently exceeded, you put the transmission system at risk of operating equipment at temperatures in excess of design when it is carrying rated load.

This circuit rating criteria attempts to achieve a balance by specifying a method for determining the maximum ambient temperature that allows for full utilization of the transmission system without experiencing frequent violations of equipment design temperatures. When selecting a maximum ambient temperature for seasonal ratings, a system may choose to either compute a temperature based on local weather station data or may use a default value.

Switches, circuit breakers and wave traps have nameplate ratings based on a 40°C (104°F) maximum ambient temperature (if designed according to ANSI/IEEE standards). The 40°C shall be used as the summer default value for this equipment along with overhead conductors. A system that elects to use the summer default value shall utilize the nameplate rating. Those

systems wanting to compute their own maximum ambient temperature shall use the procedure described in section 10.1.3.1 and must adjust the nameplate rating using the procedure described in the appropriate section for that piece of equipment. The winter default temperature is 10°C (50°F). Whether a system elects to use the winter default value or compute its own, it shall still need to determine winter ratings for overhead conductors, switches, circuit breakers and wave traps using the procedures described in the appropriate section (nameplate ratings apply for summer months only).

The maximum ambient temperature, whether based on a default value or a computed value, represents a ceiling or highest number that can be used for rating circuits. A system may rate their equipment at some lower temperature but cannot exceed the default value or the computed value. This allows systems that have historically computed their circuit ratings using an ambient temperature below the default value or the computed value to continue using this value for rating purposes.

However, for a system that has historically utilized an ambient temperature that exceeds both the default value and the computed value, it must lower the maximum ambient temperature it is using for rating purposes such that it equals either the default value or the computed value, whichever of the two the system elects to use.

10.1.2.2 Average Ambient Temperature

According to ANSI standards, both power transformers and current transformers are rated using average ambient temperature. The average temperature is calculated by averaging 24 consecutive hourly readings. The maximum daily temperature should not be more than 10°C greater than the average temperature. Power transformers and current transformers both have nameplate ratings based on a 30°C (86°F) average ambient temperature. The 30°C shall be used as the summer default value for this equipment. Using the 30°C average ambient temperature allows for a maximum daily temperature of 40°C that is consistent with the summer default value used for other equipment. The winter default value is 0°C (32°F). It allows a maximum daily temperature of 10°C (50°F) for the winter months and is consistent with the winter default value used for other equipment. A system can elect to compute its own average temperature using the procedure described in Section 10.3.1.2. Using a computed average ambient temperature (either summer or winter) for power transformers shall require adjusting its nameplate rating. The default average ambient temperature value of the transformer has been used to determine its emergency rating and an acceptable loss of life. This circuit rating criteria uses the power transformer nameplate rating as a year-round rating (both summer and winter). Consequently, even though an average ambient temperature can be computed, using it to adjust transformer nameplate ratings will affect emergency ratings and is not recommended.

Similar to the maximum ambient temperature the default or computed average ambient temperature represents a ceiling or highest number that can be used for rating power transformers and current transformers. A system can use an average ambient temperature that is less than the default or computed amount but cannot use one that exceeds the default or computed amount.

10.1.3 **PROCEDURE FOR COMPUTING AMBIENT TEMPERATURE**

Systems have the option of either using default values or computing maximum ambient temperature and average ambient temperature. This section describes the procedure for computing ambient temperatures. In general, it is intended that a system would use this procedure a single time and once they have determined their maximum and average ambient temperature, these amounts would remain constant for all future circuit ratings.

When computing ambient temperature, systems shall use temperature readings from nearest weather stations. Some control areas span a large area that may encompass several weather stations. If a large enough geographic area is involved, it is also possible that the computed ambient temperature could be significantly higher at one or more of the weather stations. For these control areas, they may choose to either divide the control area into sections using the highest ambient temperature readings from weather stations within the section or, to maintain consistent ratings throughout the control area, they may choose to use the highest ambient temperature of all weather stations within the control area.

A system is not required to use the same weather station when computing summer and winter ambient temperature. However, the weather station selected must be within the control area or, if there are no weather stations within the control area, it must be the nearest weather station to the control area. Once a weather station is selected for either summer readings or winter readings, all or the readings for each season must come from the same weather station.

10.1.3.1 Maximum Ambient Temperature

The summer maximum ambient temperature is determined by averaging the top 1% of the hourly temperature readings from the nearest weather station for the summer months (June through September). The summer average shall be computed each year for the past five years and the highest average shall be selected as the summer maximum ambient temperature.

The winter maximum ambient temperature is determined by averaging the top 1% of the hourly temperature readings from the nearest weather station for the winter months (December through February). The winter average shall be computed each year for the past five years and the highest average shall be selected as the winter maximum ambient temperature.

10.1.3.2 Average Ambient Temperature

The summer average ambient temperature is determined by averaging the 24 consecutive hourly temperature readings from the nearest weather station for the summer months (June through September). This summer average shall be computed using an average of the five hottest days during the four months. It shall be computed for each year for the past five years and the highest average shall be selected as the summer average ambient temperature.

The winter average ambient temperature shall be computed in a similar manner but shall use the average of the hottest 24 consecutive hourly temperature readings from the nearest weather station for the winter months (December through February). The winter months are defined as the three consecutive months that overlap the end of the year. The winter average

shall be computed similar to the summer average using an average of the five hottest days during the three months. The winter average shall be computed each year for the past five years and the highest average shall be selected as the winter average ambient temperature.

10.2 UNDERGROUND CABLES

The determination of the ampacity of an underground cable is dependent upon the maximum temperature the conductor can withstand without causing significant thermal deterioration. This temperature can refer to steady-state, emergency or short-circuit conditions. It is the specification of this temperature limit that limits the cable's ampacity. The Neher-McGrath procedure is the basis of steady-state ampacity calculations. This current rating can be calculated by solving the thermal equivalent of Ohm's Law and in its basic form is shown by the equation below:

$$I = \sqrt{\frac{T_c - T_a - \Delta T_d}{R_{el} \times R_{th}}} \times 10^3 A$$

where,

T_c = allowable conductor temperature, °C
 T_a = ambient earth temperature, °C
 ΔT_d = temperature rise due to dielectric loss, °C
 R_{el} = electrical resistance, $\mu\Omega/\text{ft}$.
 R_{th} = thermal resistance, TOF (thermal ohm-feet)

The basic thermal circuit consists of cable heat (in watts) that corresponds to electrical current (in amperes), thermal resistance (in thermal ohm-feet) that corresponds to electrical resistance (in ohms) and temperature drop (in °C) that corresponds to voltage drop (in volts). The electrical analogy and thermal circuit are shown below.

Simply stated, the ampacity calculation consists of the calculation of losses and the temperature rise due to those losses flowing through the various resistances. The procedure can be summarized as follows:

- 1) Determine the cable construction and conductor size of the existing cable.
- 2) Refer to AEIC specifications for the maximum allowable conductor temperature for this cable. Determine the temperature rise over ambient earth temperature that will give this value.
- 3) Calculate dielectric loss.
- 4) Calculate the electrical resistances of each current-carrying component of the system for the expected operating temperature of that component.
- 5) Calculate the thermal resistance of each component of the system, including the earth.
- 6) Calculate the temperature rise due to dielectric loss flowing through the thermal resistances, and subtract that number from the total available temperature rise.
- 7) Solve the Ohm's law equivalent circuit to determine the ampacity that achieves the allowable temperature rise.

The total thermal circuit for a self-cooled buried transmission cable is shown as an electrical analog in figure below.

The following equations summarize the calculation of the allowable current:

$$I = \sqrt{\frac{\Delta T}{\sum R_{ac} \times \overline{R_{th}}}}, (kA)$$

Temperature Rise Due to Dielectric Losses

For pipe-type cables:

$$\Delta T_d = W_d \left(\frac{1}{2} \overline{R_i} + \overline{R_{sd}} + \overline{R_{pc}} + \overline{R_{de}} + \overline{R_{dcorr}} \right), (C^\circ)$$

For single-conductor cables:

$$\Delta T_d = W_d \left(\frac{1}{2} \overline{R_i} + \overline{R_j} + \overline{R_{sd}} + \overline{R_d} + \overline{R_{pc}} + \overline{R_{de}} + \overline{R_{dcor}} \right), (C^\circ)$$

where,

- R_i = jacket thermal resistance (TOF),
- R_{sd} = thermal resistance between cable surface and surrounding enclosure (TOF),
- R_{pc} = pipe-covering thermal resistance (TOF),
- R_{dc} = d-c resistance of conductor (TOF),
- R_{dcor} = correction factor for controlled backfill or concrete-encased duct (TOF),
- R_j = thermal resistance of jacket (TOF),

Available Temperature Rise for Current-Dependent Losses

$$\Delta T_c = T_c - T_a - \Delta T_d - \Delta T_{int}, (C^\circ)$$

where,

- T_c = allowable conductor temperature (C°)
- T_a = ambient earth temperature (C°)
- ΔT_d = conductor temperature rise due to dielectric and charging-current losses (C°)
- ΔT_{int} = temperature rise due to extraneous heat source (C°)

Summation of Electrical and Thermal Resistances

For pipe-type cables:

$$\sum R_{ac} \overline{R_{th}} = R_{acc} \overline{R_i} + R_{acs} \overline{R_{sd}} + R_{acp} \overline{R_{pc}} + R_{acp} \overline{R_{dx}} + R_{acp} \overline{R_{earth}} + R_{acp} \overline{R_{mut}} + R_{acp} \overline{R_{ccorr}}, \left(\frac{^\circ C}{kA^2} \right)$$

For single-conductor cables:

$$\sum R_{ac} \overline{R_{th}} = R_{acc} \overline{R_i} + R_{acs} \overline{R_j} + R_{acs} \overline{R_{sd}} + R_{acs} \overline{R_d} + R_{acs} \overline{R_{dx}} + R_{acs} \overline{R_{earth}} + R_{acs} \overline{R_{mut}} + R_{acs} \overline{R_{ccorr}}, \left(\frac{^{\circ}C}{kA^2} \right)$$

where,

- D_e = diameter at start of the earth portion of the thermal circuit (in.)
- D_x = fictitious diameter at which the effect of loss factor commences (in.)
- R_{acc} = conductor ac resistance ($\mu\Omega/\text{ft.}$)
- R_{acp} = ac resistance of pipe ($\mu\Omega/\text{ft.}$)
- R_{acs} = ac resistance of shield ($\mu\Omega/\text{ft.}$)
- R_{ccorr} = correction factor, current-dependent losses (TOF)
- R_{dx} = thermal resistance from D_e to diameter D_x (TOF)
- R_{earth} = thermal resistance from D_x to diameter earth (TOF)
- R_{mut} = mutual-heating earth thermal resistance term (TOF)

For further and more detailed information see the [EPRI Underground Transmission Systems Reference Book](#), Chapter 5: Ampacity. The above equations apply to buried cable installations. For above ground installation (e.g. bridge crossings, tunnel installations or riser sections), there are three main differences. First, solar radiation provides heat input. The Neher-McGrath method does not consider this factor. Second, heat transfer by conduction is negligible. And third, heat transfer for cables in air is by free or forced convection and by radiation. Generally, the ampacity of an identical cable circuit installed in the air will be greater than that for a buried cable circuit. The equations will not be presented here and may be found in the [EPRI Underground Transmission Systems Reference Book](#), Chapter 5: Ampacity.

Under emergency conditions it may become necessary to exceed the normal temperature limit of the cable. A new cable rating may be calculated by substituting the new temperature limit into the equations seen above. The key to calculating the emergency rating will be establishing the allowable temperature increase. One must be careful not to overestimate the ability of the cable to withstand higher temperatures as well as not to underestimate the loss of life that will occur with operation above normal temperatures.

10.3 SWITCHES

Switch ratings are discussed in ANSI/IEEE C37.37 **Loading Guide for AC High-Voltage Air Switches (in excess of 1000 volts)** and in ANSI/IEEE C37.37a **Loading Guide for AC High-Voltage Air Switch Under Emergency conditions**. In general, the allowable continuous current of a switch is based on the allowable maximum temperature of the switch parts and is affected by the ambient temperature. Provisions are made for loading the switches under emergency conditions in the above referenced ANSI/IEEE Standard C37.37a. Therefore both Normal and Four-Hour Emergency circuit ratings will be developed based upon the allowable continuous current capability rating of the switch. These ratings will be treated as the loading limits for normal and emergency conditions.

A switch is made up of a number of different switch parts, which are classified and grouped in accordance with their material and function into switch part classes, each of which is given a switch part class designation of the form xO#. Examples of the switch part class designations are A01, D06, J010, N01, and W09. The loadability factors of each switch part class, as a function of ambient temperature, are represented by a curve. Tabular representations of these curves for normal continuous load conditions are contained in Table C-3 for non-enclosed indoor and outdoor switches and in Table C-4 for enclosed indoor and outdoor switches. Tabular representations of these curves for emergency load conditions with a maximum emergency loading duration of four hours are contained in Table C-5 for non-enclosed indoor and outdoor switches and in Table C-6 for enclosed indoor and outdoor switches.

The Allowable Continuous Current Class (ACCC) designation of a switch is a code which identifies a composite loadability curve made up from the limiting switch part classes. In most instances, the ACCC designation will be contained on the switch's nameplate. However, air switches designed in accordance with ANSI C37.30- and earlier may not have an ACCC designation on the nameplate. Such non-enclosed switches having a 30°C limit of observable temperature rise in a maximum ambient of 40°C, indicative of an allowable maximum temperature of 70°C, are assigned an ACCC designation of A01. In like manner, such enclosed switches having a 30°C limit of observable temperature rise in a maximum ambient of 55°C, indicative of an allowable maximum temperature of 85°C, are assigned an ACCC designation of N01.

ACCC designations were developed such that the first character, x, designates that the specific normal loadability factors for that class are represented by Curve x when the ambient is between 60°C and 25°C.; the second character in the ACCC designation, O, represents the specific normal loadability factor for that class at 25°C. (All curves intersect at O and have a loadability factor of 1.22 at 25°C ambient); and the third character, #, designates that the specific normal loadability factors for that class are represented by Curve# when the ambient is between 25°C and -30°C.

In some instances, a composite curve must be developed due to the fact that the switch is constructed such that some parts of the switch have different switch part class designations than other parts of the switch. In this situation, the switch's ACCC designation will differ from the standard ACCC designations listed in the tables and instead will be a composite of two of the standard ACCC designations. An example of this would be a switch having a designation of D06, signifying that the switch is constructed using some parts having the switch part class designations of D04 and some parts having the switch part class designations of F06.

Based upon the ACCC designation of a switch, a normal loadability factor curve can be developed for the switch in the following manner.

- 1) Using Table C-2 for non-enclosed indoor and outdoor switches or Table C-3 for enclosed indoor and outdoor switches, the appropriate values for Curve x, Curve O, and Curve# are determined based upon the switch's ACCC designation. These curves will be joined together to form a composite normal loadability factor curve for the switch for the ambient temperature range from 60°C to -30°C.

- 2) The normal loadability factor for specific ambient temperatures can be determined from the composite loadability factor curve. If the normal loadability factor for a specific temperature, which is not represented in the composite table, is desired, the normal loadability factor may be calculated by interpolating between the nearest known values. In no case shall the normal loadability factor exceed 2.00.
- 3) The Allowable Continuous Current Capability of a switch at a given ambient temperature is equal to the normal loadability factor at that ambient temperature multiplied by rated continuous current of the switch, which is contained on the switch's nameplate.

The Allowable Continuous Current Capability of a switch at a given ambient temperature will be treated as the normal ratings of the switch for that ambient temperature.

Example: The following example should serve to clarify this process:

Problem:

- 1) Develop a loadability factor table for a 1200-amp non-enclosed switch with ACCC designation D06.
- 2) Using this table, determine the Allowable Continuous Current Capability of this switch at an ambient temperature of 0°C.

Solution:

- 1) The normal loadability factor table, Table C-1, was created using Table C-3 for a non-enclosed switch with ACCC designation 006. For the temperatures from 25°C to 60°C, the loadability factors from Curve D were used. The standard loadability factor of 1.22 was used for an ambient of 25°C. The loadability factors from Curve 6 were used for the temperatures from 20°C to -30°C.
- 2) Based upon Table C-1, the Allowable Continuous Current Capability of a 1200-A non-enclosed switch designated D06 at 0°C would be:

Loadability Factor @ 0°C * Rated Current

or

$1.41 * 1200 \text{ amps} = 1692 \text{ amps}$

For operation at 0°C, the normal rating of switch would be 1692 amps.

Maximum Ambient Temperature	Loadability Factor	Curve Used
60°C	.84	Curve D
50°C	.96	
40°C	1.07	
30°C	1.18	
25°C	1.22	Standard
20°C	1.27	Curve 6
10°C	1.34	
0°C	1.41	
-10°C	1.47	
-20°C	1.54	
-30°C	1.60	

Table C-1: Normal Loadability Factors for a Type D06 Switch

The Four-Hour Emergency Load Current carrying capability of a switch can be determined based upon the switch's ACCC designation. A four-hour emergency loadability factor curve can be developed for the switch in the following manner.

- 1) Using Table C-5 for non-enclosed indoor and outdoor switches or Table C-6 for enclosed indoor and outdoor switches, the appropriate emergency loadability factors are determined based upon the switch's ACCC designation. In developing the composite curve, the emergency loadability factor curves of the two standard ACCC designations must be examined and the most limiting loadability factor at each ambient temperature will be used to form a composite emergency loadability factor curve for the switch for the ambient temperature range from 60°C to -30°C.
- 2) The emergency loadability factor for specific ambient temperatures can be determined from the composite loadability factor curve. If the emergency loadability factor for a specific temperature, which is not represented in the composite table, is desired, the emergency loadability factor may be calculated by interpolating between the nearest known values. In no case shall the emergency loadability factor exceed 2.00.
- 3) The Four-Hour Emergency Load Current Capability of a switch at a given ambient temperature is equal to the emergency loadability factor at that ambient temperature multiplied by rated continuous current of the switch, which is contained on the switch's nameplate.

The Four-Hour Emergency Load Current Capability of a switch at a given ambient temperature will be treated as the Emergency ratings of the switch for that ambient temperature.

Example: The following example should serve to clarify this process

Problem:

- 1) Develop an emergency loadability factor table for a 1200-amp non-enclosed switch with ACCC designation D06.
- 2) Using this table, determine the Four-Hour Emergency Load Current Capability of this switch at an ambient temperature of 0°C.

Solution:

- 1) The emergency loadability factor table, Table C-2, was created using Table C-5 for a non-enclosed switch with ACCC designation D06. The composite curve was developed by examining the emergency loadability curves for switches having ACCC designations of D04 and F06. At each of the ambient temperatures, the most limiting loadability factor was chosen from the D04 and F06 curves to form the composite curve.
- 2) Based upon Table C- 3, the Four-Hour Emergency Load Current Capability of a 1200-A non-enclosed switch designated D06 at 0°C would be:

Emergency Loadability Factor@ 0°C * Rated Current

or

$$1.54 * 1200 \text{ amps} = 1848 \text{ amps}$$

For operation at 0°C, the Four-Hour Emergency ratings of the switch would be 1848 amps.

Maximum Ambient Temperature	Loadability Factor Curve 004	Loadability Factor Curve F06	Composite Curve for a D06 Switch
60°C	1.08	1.11	1.08
50°C	1.18	1.19	1.18
40°C	1.28	1.27	1.27
35°C	1.32	1.30	1.30
30°C	1.36	1.34	1.34
25°C	1.41	1.37	1.37
20°C	1.45	1.41	1.41
10°C	1.53	1.47	1.47
0°C	1.60	1.54	1.54
-10°C	1.67	1.60	1.60
-20°C	1.74	1.65	1.65
-30°C	1.81	1.71	1.71

Table C-2: Four-Hour Emergency Loadability Factors for a Type D06 Switch

Maximum Ambient Temperature		ACCC Designation																					
		A01		B02		C03		D04		E05		F06		G07		H08		I09		J010		K011	
°C	°F																						
60	140	0.58	Curve A	0.67	Curve B	0.74	Curve C	0.84	Curve D	0.86	Curve E	0.92	Curve F	0.94	Curve G	0.98	Curve H	1.03	Curve I	1.06	Curve J	1.10	Curve K
50	122	0.82		0.87		0.90		0.96		0.98		1.02		1.03		1.05		1.09		1.12		1.14	
40	104	1.00		1.03		1.04		1.07		1.08		1.11		1.11		1.13		1.15		1.16		1.18	
30	86	1.15		1.17		1.17		1.18		1.18		1.19		1.19		1.20		1.20		1.20		1.21	
25	77	1.22		1.22		1.22		1.22		1.22		1.22		1.22		1.22		1.22		1.22		1.22	
20	68	1.29	Curve 1	1.29	Curve 2	1.28	Curve 3	1.28	Curve 4	1.27	Curve 5	1.27	Curve 6	1.27	Curve 7	1.26	Curve 8	1.25	Curve 9	1.24	Curve 10	1.24	Curve 11
10	50	1.41		1.40		1.38		1.36		1.34		1.34		1.32		1.32		1.30		1.28		1.28	
0	32	1.53		1.51		1.47		1.45		1.42		1.41		1.39		1.38		1.34		1.32		1.31	
-10	14	1.63		1.60		1.56		1.52		1.49		1.47		1.45		1.44		1.39		1.36		1.34	
-20	-4	1.73		1.70		1.64		1.60		1.56		1.54		1.51		1.49		1.44		1.40		1.37	
-30	-22	1.83		1.78		1.72		1.67		1.63		1.60		1.57		1.54		1.48		1.44		1.40	

Table C-3: Non-enclosed Indoor and Outdoor Switches
Switches Part Normal Loadability Factors (LF) for Various Ambient Temperatures

Maximum Ambient Temperature		ACCC Designation																					
		Q033		PO22		NO1		RO4		SO5		TO6		UO7		VO8		WO9		XO10		YO11	
°C	°F																						
60	140	0.00	Curve Q	0.45	Curve P	0.58	Curve N	0.67	Curve R	0.74	Curve S	0.84	Curve T	0.86	Curve U	0.92	Curve V	1.00	Curve W	1.05	Curve X	1.09	Curve Y
50	122	0.71		0.77		0.82		0.87		0.90		0.96		0.98		1.02		1.07		1.10		1.13	
40	104	1.00				1.00				1.00		1.00		1.03		1.04		1.08		1.11		1.13	
30	86	1.22	Curve 3	1.18	Curve 2	1.15		1.16		1.16		1.18		1.18		1.19		0.20		1.20		1.20	
25	77	1.32		1.26		1.22		1.22		1.22		1.22		1.22		1.22		1.22		1.22			
20	68	1.41		1.34		1.29	Curve 1	1.29	Curve 4	1.28	Curve 5	1.28	Curve 6	1.26	Curve 7	1.26	Curve 8	1.25	Curve 9	1.25	Curve 10	1.24	Curve 11
10	50	1.58		1.48		1.41		1.40		1.38		1.36		1.35		1.34		1.31		1.29		1.28	
0	32	1.73		1.61		1.53		1.51		1.47		1.45		1.42		1.41		1.36		1.34		1.31	
-10	14	1.87		1.73		1.63		1.61		1.56		1.53		1.50		1.47		1.41		1.40		1.35	
-20	-4	2.00		1.84		1.73		1.70		1.64		1.60		1.56		1.54		1.46		1.42		1.38	
-30	-22	2.00		1.95		1.83		1.78		1.72		1.67		1.63		1.60		1.51		1.46		1.41	

Table C-4: Enclosed Indoor and Outdoor Switches
Switches Part Normal Loadability Factors (LF) for Various Ambient Temperatures

Maximum Ambient Temperature		ACCC Designation										
		A01	B02	C03	D04	E05	F06	G07	H08	I09	J010	K011
°C	°F											
60	140	1.00	1.03	1.04	1.08	1.08	1.11	1.11	1.13	1.15	1.16	1.18
50	122	1.15	1.17	1.16	1.18	1.18	1.09	1.18	1.20	1.20	1.20	1.21
40	104	1.29	1.29	1.27	1.28	1.26	1.27	1.26	1.26	1.25	1.24	1.24
35	95	1.35	1.35	1.33	1.32	1.30	1.30	1.29	1.29	1.27	1.26	1.26
30	86	1.41	1.40	1.38	1.36	1.34	1.34	1.32	1.32	1.30	1.28	1.28
25	77	1.47	1.46	1.42	1.41	1.38	1.37	1.36	1.35	1.32	1.30	1.29
20	68	1.53	1.51	1.47	1.45	1.42	1.41	1.39	1.38	1.35	1.32	1.31
10	50	1.63	1.361	1.56	1.53	1.49	1.47	1.45	1.44	1.39	1.36	1.34
0	32	1.73	1.70	1.64	1.60	1.56	1.54	1.51	1.49	1.44	1.40	1.37
-10	14	1.83	1.78	1.72	1.67	1.63	1.60	1.57	1.54	1.48	1.44	1.40
-20	-4	1.92	1.87	1.80	1.74	1.69	1.65	1.62	1.59	1.52	1.47	1.43
-30	-22	2.00	1.95	1.87	1.80	1.76	1.71	1.68	1.64	1.56	1.51	1.46

Table C-5: Non-enclosed Indoor Outdoor Switches
Switch Part Emergency Loadability Factors (ELF) for Various Ambient Temperatures

Maximum Ambient Temperature		ACCC Designation										
		Q03	PO2	NO1	RO4	SO5	TO6	UO7	VO8	WO9	XO10	YO11
°C	°F											
60	140	1.00	1.00	1.00	1.03	1.04	1.08	1.08	1.11	1.13	1.15	1.17
50	122	1.22	1.18	1.15	1.17	1.16	1.18	1.18	1.19	1.20	1.20	1.24
40	104	1.41	1.34	1.29	1.29	1.27	1.28	1.26	1.27	1.25	1.25	1.24
35	95	1.50	1.41	1.35	1.35	1.33	1.32	1.30	1.30	1.28	1.27	1.26
30	86	1.58	1.48	1.35	1.35	1.33	1.32	1.30	1.30	1.28	1.27	1.26
25	77	1.66	1.55	1.47	1.46	1.42	1.41	1.38	1.37	1.34	1.31	1.30
20	68	1.73	1.61	1.53	1.51	1.47	1.45	1.42	1.41	1.36	1.33	1.31
10	50	1.87	1.73	1.63	1.61	1.56	1.53	1.49	1.47	1.41	1.38	1.35
0	32	2.00	1.84	1.73	1.70	1.64	1.60	1.56	1.54	1.46	1.42	1.38
-10	14	2.00	1.95	1.83	1.78	1.72	1.67	1.63	1.60	1.51	1.46	1.41
-20	-4	2.00	2.00	1.92	1.87	1.80	1.74	1.69	1.65	1.56	1.50	1.44
-30	-22	2.00	2.00	2.00	1.95	1.87	1.80	1.76	1.71	1.60	1.54	1.47

Table C-6: Enclosed Indoor Outdoor Switches
Switch Part Emergency Loadability Factors (ELF) for Various Ambient Temperatures

10.4 WAVE TRAPS

Wave trap ratings are discussed in several sources. The ratings for the older wave traps in which the main coil is designed as a single-phase, air-cored inductor of the dry type are discussed in ANSI C93.3-1981, **Requirements for Power-Line Carrier Line Traps** and in Westinghouse Electric's Curve No. 511545. The ratings of the newer type of wave traps consisting of a coil of wire encapsulated in an epoxy resin are not covered in an ANSI/IEEE standard. The ratings for this type of wave trap were developed based upon guidelines supplied by the manufacturer, Trench Electric. In general, the rated continuous current for wave traps is based on the maximum permissible temperature rise limitations of the wave trap when it is carrying rated current at an ambient temperature of 40°C. The total temperature of the wave trap under service conditions depends both on the actual load current and the actual ambient temperature. It is, therefore, possible to operate at a current higher than rated continuous current when ambient temperature is less than 40°C, provided that the allowable total temperature rise limitation is not exceeded.

Ambient-compensated continuous current ratings will be used for the normal seasonal circuit ratings of both types of wave traps. These seasonal ratings will be based on the nameplate rating of the wave trap with an adjustment made for changes in the rated continuous current due to changes in the ambient air temperature. The ambient-compensated normal loadability factors for Dry-Type, Air-Cored Inductor wave traps are given in Table D-1 and were obtained from Westinghouse Electric's Curve No. 511545. The ambient-compensated loadability factors for Epoxy-Encapsulated Inductor wave traps were developed from Trench Electric's loadability curves and are given in Table D-2. To determine the normal seasonal circuit rating, simply multiply the wave trap's nameplate rating by the appropriate normal loadability factor. Interpolation may be used to obtain loadability factors at ambients other than those specified in the tables.

Normal life expectancy will occur with a wave trap operated at or below its ambient-compensated normal current rating. Any value of currents in excess of the ambient-compensated normal current ratings may cause the designed temperature rise to be exceeded and may shorten the life expectancy of the wave trap. For wave traps utilizing an air-cored inductor, an acceptable level of emergency overload current has been determined and is specified in ANSI Standard C93.3-1981, **Requirements for Power-Line Carrier Line Traps**. Ambient-compensated emergency current ratings will be used for the emergency seasonal circuit ratings of wave traps utilizing an air-cored inductor. These seasonal ratings will be based on the Four-Hour Emergency Overload Current-Carrying capability of the wave trap with an adjustment made for changes in the continuous current due to changes in the ambient air temperature. The wave traps may be loaded to this emergency level for a maximum of four hours per cycle. Before an emergency load cycle, the wave trap loading must be at or below the normal seasonal loading level for at least two hours.

The emergency seasonal circuit rating can be developed by simply multiplying the wave trap's nameplate rating by the appropriate emergency loadability factor. The ambient-compensated emergency loadability factors for Dry-Type, Air-Cored Inductor wave traps are given in Table D-1 and were obtained from ANSI Standard C93.3-1981, **Requirements for Power-Line Carrier Line Traps**. Interpolation may be used to obtain loadability factors at ambients other than those specified in the tables.

Average Ambient Air Temperature	Loadability Factors	
	Normal	4-Hour Emergency
40°C	1.00	1.10
20°C	1.05	1.15
0°C	1.10	1.20
-20°C	1.10	1.25

Table D-1: Loadability Factors for Dry-Type Air-Cored Inductor Wave Traps

Average Ambient Air Temperature	Loadability Factors	
	Normal	4-Hour Emergency
40°C	1.00	N/A
20°C	1.13	N/A
0°C	1.25	N/A
-20°C	1.36	N/A

Table D-2: Loadability Factors for Epoxy-Encapsulated Inductor Wave Traps

Trench Electric, the manufacturer of the newer type of wave traps which consist of a coil of wire encapsulated in an epoxy resin, has stated that no overload capability is available on this type of wave trap. For this reason, no separate emergency rating will be assigned to this type of wave trap and the ambient-compensated current rating will be used for both the circuit's normal and emergency ratings.

July 16, 1985

Westinghouse Electric Corporation

Subject: Line Trap Loadings

Attached is one copy of the Appendix to ANSI Standard C93.3-1981 titled "Guide for Emergency Overload Current Capability of Line Traps".

This guide is current and, all Westinghouse traps built since 1970 conforms to the overload capability designated therein.

The Curve No. 511545 supplied applies to Westinghouse Line Traps built prior to 1970. Note that the overload capability of traps 2000 amperes and below on this curve are the same as that specified in the Standard. The larger traps were modified in design in order to conform to the Standard requirement.

July 25, 1985

Trench Electric

Attached are three copies of the overload current U.S. temperature curves. General Electric did not send any such information when we purchased their Line Trap division but the curves should be approximately the same.

10.5 CURRENT TRANSFORMERS

Current transformer ratings are discussed in ANSI/IEEE C57.13 **IEEE Standard Requirements for Instrument Transformers** and in Westinghouse Electric's technical paper titled "**Memorandum on Thermal Current Characteristics of Current Transformers Used With Power Circuit Breakers and Power Transformers**". In general, the rated continuous current for current transformers is based on the maximum permissible temperature limitations of the various parts of the current transformer when it is carrying rated current at a 24-hour average ambient temperature of 30°C (with a maximum hourly temperature of 40°C). The total temperature of these parts under service conditions depends both on the actual load current, the actual average ambient temperature, and the service environment of the current transformer. Depending on the environment, it is, therefore, possible to operate at a current higher than rated continuous current when the average ambient temperature is less than 30 °C, provided that the allowable total temperature limit is not exceeded. Similarly, when the average ambient temperature is greater than 30°C, the current must be reduced to less than rated continuous current to keep total temperatures within allowable limits.

The environment in which a current transformer operates has a large effect on the current-carrying capability of the current transformer. Separately mounted current transformers have been assigned rated primary and secondary currents and continuous-thermal-current rating factors by the manufacturer. Separately mounted current transformers are designed to meet these characteristics by the independent control of such parameters as: current density in the primary and secondary windings, geometry, area of radiating surfaces, and heat transfer characteristics. Permissible loading as a function of ambient temperatures and continuous-thermal-current rating factors, and permissible overloading under emergency conditions, is covered in **ANSI/IEEE C57.13 IEEE Standard Requirements for Instrument Transformers**.

Bushing current transformers mounted in power circuit breakers and power transformers differ from separately mounted current transformers in that the design parameters cannot be independently controlled since they are restricted by the characteristics of the power apparatus in which they are mounted. Bushing current transformers, when attached to various power apparatus, are subjected to wide variations in their environmental ambient temperature. This variation is dependent upon the thermal characteristics of the power apparatus and the relative current loading with respect to the rated current of the power apparatus and its bushing current transformer. For this reason, continuous-thermal-current rating factors are not typically calculated by the manufacturer for bushing current transformers. Instead, the continuous-thermal-current rating factors must be determined for each application. Permissible loading and permissible overloading of bushing current transformers is covered in Westinghouse Electric's technical paper titled "**Memorandum on Thermal Current Characteristics of Current Transformers Used With Power Circuit Breakers and Power Transformers**".

Rating Separately Mounted Current Transformers: The maximum continuous-thermal-current ratings for separately-mounted current transformers designed for 55°C temperature rise above 30°C average ambient air temperature will vary according to actual ambient cooling air temperature. The adjusted continuous current ratings will be based on three items:

- 1) The nameplate primary and secondary current ratings at the ratio setting being used.
- 2) The average cooling air temperature at which the rating is being calculated. Note: The maximum cooling air temperature must not exceed the average cooling air temperature by more than 10°C.

- 3) The ambient-adjusted Continuous-Thermal-Current rating factor from Table E-1. The value used from this table is based upon the nameplate Continuous-Thermal-Current rating factor of the current transformer and the average ambient air temperature.

Average Cooling Temperature	Nameplate Continuous Thermal Current Rating Factor					
	1.00	1.33	1.50	2.00	3.00	4.00
60°C	0.65	0.85	1.00	1.35	2.00	2.70
50°C	0.80	1.05	1.20	1.60	2.40	3.20
40°C	0.90	1.20	1.35	1.80	2.70	3.65
30°C	1.00	1.33	1.50	2.00	3.00	4.00
20°C	1.05	1.45	1.65	2.20	3.30	4.35
10°C	1.20	1.55	1.75	2.35	3.50	4.70
0°C	1.30	1.65	1.85	2.45	3.70	5.00

Table E-1: Ambient-Adjusted Continuous-Thermal-Current Rating Factors

The ambient-adjusted continuous-thermal-current rating of a separately-mounted current transformer is equal to the ambient adjusted continuous-thermal-current rating factor multiplied by rated primary current of the current transformer at the ratio specified. According to ANSI/IEEE C57.13-1978, IEEE Standard Requirements for Instrument Transformers, no overload capability is available on separately-mounted current transformers. For this reason, normal rating of the separately-mounted current transformer will be used for both the normal seasonal circuit rating and the emergency seasonal circuit rating.

Example: The following example should serve to clarify this process:

Problem:

- 1) Determine the Summer Seasonal normal and emergency ratings for a separately-mounted current Transformer operating on a 2000:5 amp ratio with a nameplate Continuous-Thermal-Current rating of 1.50. Assume the average ambient air temperature is 30°C with a maximum temperature of 40°C for the summer period.
- 2) Determine the Winter Seasonal normal and emergency ratings for this same separately-mounted current transformer assuming an average ambient air temperature of 0°C with a maximum temperature of 10°C for the winter period.

Solution

- 1) From Table E-1, the 30°C ambient-adjusted Continuous-Thermal-Current rating factor for a separately-mounted current transformer with a nameplate Continuous-Thermal-Current rating factor of 1.50 is 1.50 (They are the same due to the fact that the nameplate C-T-C rating factor is developed at an average ambient of 30°C). The normal and emergency summer ratings would then be equal to:

Ambient-Adjusted C-T-C R.F. @ 30°C • Rated Primary Current of CT Ratio or

$$1.50 \bullet 2000 \text{ amps} = 3000 \text{ amps}$$

For operation at 30°C, both the Summer Normal and Emergency ratings of current transformer would be 3000 amps. Note: When the primary current is 3000 amps, the secondary current would be 7.5 amps due to the 2000:5 ratio used in the current transformer.

- 2) From Table E-1, the 0°C ambient-adjusted Continuous-Thermal-Current rating factor for a separately-mounted current transformer with a nameplate Continuous-Thermal-Current rating factor of 1.50 is 1.85. The normal and emergency summer ratings would then be equal to:

Ambient-Adjusted C-T-C R.F. @ 0°C *Rated Primary Current of CT Ratio or

$$1.85 * 2000 \text{ amps} = 3700 \text{ amps}$$

For operation at 0°C, both the Winter Normal and Emergency ratings of current transformer would be 3700 amps. Note: When the primary current is 3700 amps, the secondary current would be 9.25 amps due to the 2000:5 ratio used in the current transformer.

Rating Bushing Current Transformers Mounted in Power Apparatus: The continuous current rating of current transformers mounted in power circuit breakers or power transformers is determined as follows:

- 1) When the bushing current transformer ratio being used has the same primary current rating as the breaker continuous current rating or the power transformer rated current, the continuous current rating factor of the current transformer is 1.0 (unity).
- 2) When the primary current rating of the ratio being used on the current transformer is greater than the continuous current rating of the breaker or greater than the power transformer rated current, then the maximum load current is limited by the breaker or power transformer rating, whichever is applicable.
- 3) When the primary current rating of the ratio being used is less than the continuous current rating of the breaker or the power transformer rated current, then the maximum load current is limited by the continuous thermal current rating of the current transformer when operating on that ratio. Under these conditions, the temperature rises of the current transformer and the power apparatus would be lower, and therefore, the current transformer can be operated at a continuous thermal rating factor greater than 1.0. In this situation, the permissible continuous -thermal-current rating factor is calculated based upon constant maximum power dissipation, using the following equation:

$$R.F. = \sqrt{\frac{I_{pa}}{I_{ct}}}$$

where,

R.F. =Continuous-thermal-current rating factor

I_{pa} = Power apparatus continuous current rating (amps)
 I_{ct} = Primary current rating of bushing current transformer ratio used (amps)

The results calculated from this equation should be so limited that the maximum rating factor does not exceed 2.0 and that the continuous current rating of the breaker or the rated current of the power transformer is not exceeded.

Note: The situation, in which the primary current rating of the current transformer ratio being used is less than the continuous current rating of the breaker or the power transformer rated current, is an unusual case and should be viewed as the exception rather than the rule.

The normal rating of a bushing current transformer is equal to the continuous-thermal-current rating factor multiplied by rated primary current of the current transformer at the ratio specified.

Some short-term emergency overload capability is typically available on bushing current transformers mounted on power circuit breakers and power transformers. However, when utilizing this emergency overload capability, care must be taken to coordinate the loading limit of the current transformer with the overall application limitations of the other equipment affected by the current transformer loading. Particular care should be taken to avoid exceeding the maximum metering capability of metering equipment attached to the current transformer secondary.

10.6 CIRCUIT BREAKERS

Circuit breaker ratings are discussed in ANSI/IEEE C37.010, **IEEE Application Guide for AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis** and in ANSI/IEEE C37.010b, **IEEE Standard for Emergency Load Current-Carrying Capability**. These standards discuss the development of ratings for circuit breakers from the standpoint of the manufacturers. From a practical standpoint, the application of the methods discussed in these standards to circuit breakers in operation will be very difficult, due to the fact that this application would require contacting the manufacturer for detailed design information for each circuit breaker being rated. Therefore, it is recommended that for any circuit breakers which do not limit flows on the transmission system, the nameplate rated continuous current be used as the circuit breaker's normal and emergency circuit ratings. For circuit breakers which limit flows on the transmission system, the manufacturer could be consulted and ambient-compensated normal and emergency ratings could be developed for the circuit breaker using the methods described in the ANSI/IEEE Standards. These methods are described in the remainder of this section.

In general, the rated continuous current for circuit breakers is based on the maximum permissible temperature limitations of the various parts of the circuit breaker when it is carrying rated current at an ambient temperature of 40°C. The total temperature of these parts under service conditions depends both on the actual load current and the actual ambient temperature. It is, therefore, possible to operate at a current higher than rated continuous current when ambient temperature is less than 40°C, provided that the allowable total temperature limit is not exceeded. Similarly, when the ambient temperature is greater than 40°C, the current must be reduced to less than rated continuous current to keep total temperatures within allowable limits.

Ambient-compensated continuous current ratings will be used for the normal seasonal circuit ratings of outdoor circuit breakers. These seasonal ratings will be based on the nameplate rating of the circuit breaker with an adjustment made for changes in the rated continuous current due to

changes in the ambient air temperature. The seasonal loading limit shall be coordinated with the overall application limitations of the other equipment affected by the loading of the circuit breaker, such as cables and current transformers. (See other appropriate appendices to determine limitations on the other equipment.) For circuit breakers used in metal-clad switchgear, no seasonal ratings will be calculated. Instead, the loading limit shall be coordinated with the overall application limitations of the switchgear.

The seasonal normal circuit breaker ratings can be determined in the following manner:

- 1) A circuit breaker's rated continuous current is based on the maximum permissible total temperature limitations, θ_{max} , and θ_r , of the various parts of the circuit breaker when it is carrying rated current at an ambient temperature of 40°C. The constructional features of a circuit breaker dictate the appropriate values of θ_{max} and θ_r , to be used in the calculation of ambient-compensated ratings. The major components of a circuit breaker have several different temperature limitations which are shown in Table F-1. In order that none of these temperature limitations be exceeded when the load current is adjusted to the value permitted by the actual ambient temperature, the values of θ_{max} and θ_r , should be determined as follows:
 - (a) If the actual temperature is less than 40°C, the applicable component with the highest specified values of allowable temperature limitations should be used for determining θ_{max} and θ_r ,
 - (b) If the actual temperature is greater than 40°C, the applicable component with the lowest specified values of allowable temperature limitations should be used for determining θ_{max} and θ_r ,

Component Description	Limit of Temperature Rise, °C θ_r	Limit of Total Temperature, °C θ_t
Circuit breaker parts handled by the operator in the normal course of his duties	10	50
Copper contacts, copper-to-copper conducting joints, external surfaces accessible to the operator in the normal course of his duties, external terminal connected to bushing	30	70
Top oil	40	80
Breaker terminal to be connected to 85°C	45	85
Hottest spot temperature of parts where they contact oil; silver (or equal) contacts in oil; silver (or equal) conduction joints in oil	50	90
Silver (or equal) contact in air; silver (or equal) conducting joints in air; hottest spot of bushing conductor or of bushing metal parts in contact with Class A insulation or with oil; hottest spot winding temperature 55°C rise of current transformers	60	105
External surfaces not accessible to an operator in the normal course of his duties	70	110
Hottest spot winding temperature of 80°C dry type current transformers	110	150

Table F-1 Summary of Temperature Limitations for Circuit Breaker Components

- 2) The continuous current which a circuit breaker can carry at a given ambient temperature without exceeding its total temperature limitations is given by the formula:

$$I_a = I_r \left(\frac{\theta_{max} - \theta_a}{\theta_r} \right)$$

where,

I_a = allowable continuous load current, in amperes, at the actual ambient temperature

θ_a (I_a is not to exceed two times I_r)

I_r = rated continuous current, in amperes

θ_{max} = allowable hottest spot total temperature ($\theta_{max} = \theta_r + 40^\circ\text{C}$), in degrees Celsius

θ_a = actual ambient temperature expected (between -30°C and 60°C), in degrees Celsius

θ_r = allowable hottest spot temperature rise at rated current, in degrees Celsius

- 3) The ambient-compensated allowable continuous load current, I_a , calculated in Step 2 will be treated as the seasonal normal rating of the circuit breaker.

Normal life expectancy will occur with a circuit breaker operated at continuous nameplate rating. When operated for one or more load cycles above nameplate rating, the material used in manufacturing the circuit breaker deteriorates at a faster rate than normal. The rate of deterioration is commonly expressed as a percentage loss of life. According to ANSI/IEEE C37.010b- **IEEE Standard for Emergency Load Current-Carrying Capability**, a circuit breaker can experience operation at 10°C above the limits of total temperature for which the circuit breaker was designed for up to two periods of an emergency cycle of eight hours before an unreasonable loss of life occurs and maintenance is required. Inspection and maintenance must be performed in accordance with manufacturer's recommendations before the circuit breaker can be subjected to additional emergency cycles. In addition, following each emergency cycle, the load current shall be limited to 95% of the rated continuous current as modified by ambient compensation for a minimum of two hours.

Ambient-compensated emergency current ratings will be used for the emergency seasonal circuit ratings of outdoor circuit breakers. These seasonal ratings will be based on the Eight-Hour Emergency Load Current-Carrying capability of the circuit breaker with an adjustment made for changes in the continuous current due to changes in the ambient air temperature. The seasonal emergency loading limit shall be coordinated with the overall application limitations of the other equipment affected by the loading of the circuit breaker, such as cables and current transformers.

The seasonal emergency circuit breaker ratings can be determined in the following manner:

- 1) Emergency load current-carrying capability factors with ambient temperatures at 40°C are listed in Table F-2 for each limiting temperature of various circuit breaker components. The factors have been selected to allow operation at 10°C above the limits of total temperature for an emergency period of eight hours. The factors are expressed as the ratio of emergency load current allowed at an ambient temperature of 40°C, I_e , to the rated continuous current, I_r , and can be applied with the following restrictions:
 - (a) The circuit breaker component with the highest values of limiting temperature, θ_{max} , and θ_r , shall be used to select the proper emergency load current-carrying capability factor from Table F- 2.
 - (b) The Eight-Hour factor shall be used for a cycle of operation consisting of separate periods of no longer than eight hours each, with not more than two such occurrences before maintenance.
 - (c) Each cycle of operation is separate, and no time-current integration is permissible to increase the number of periods within a given maintenance cycle.

Limiting Temperatures of (°C) of Different Breaker Components							
θ_{max}	70	80	85	90	105	110	150
θ_r	30	40	45	50	65	70	110
	1.17	1.13	1.11	1.10	1.08	1.07	1.05

Table F-2: Emergency Load Current. Carrying Capability Factor (I_e/I_r)

- 2) The emergency load current, I_{ea} of a circuit breaker operating at an ambient temperature other than 40°C can be calculated by this equation:

$$I_{ea} = I_r \left(\frac{\theta_{max} - \theta_a}{\theta_r} + \left(\frac{I_e}{I_r} \right)^{1.8} - 1 \right)^{\frac{1}{1.8}}$$

where

I_{ea} = emergency load current, in amperes, at the actual ambient temperature θ_a (I_{ea} is not to exceed two times I_r)

I_r = rated continuous current in amperes

I_e = emergency load current, in amperes, at 40°C ambient temperature

θ_a = actual ambient temperature expected (between -30°C and 60°C) in degrees Celsius

θ_r = allowable hottest spot temperature rise at rated current, in degrees Celsius

I_e/I_r = emergency load current-carrying capability factor from Table F-2

- 3) The ambient-compensated emergency load current, I_{ea} calculated in Step 2 will be treated as the seasonal emergency rating of the circuit breaker.

11. GENERATOR REPORTING FORMS

Southwest Power Pool

Generator Owner/Operator Excitation System Summary Report

Instructions: Generator Owner/Operator shall fill out summary information for all units under your control concerning the number of hours the Excitation System was not operated in Automatic Voltage Control Mode.

Generator Owner/Operator	
Month Reporting Data	
Contact Name	
Contact Phone	

[illegible]

Dev. February, 2001
By SPP GWG

Generator Unit Excitation System Status Report

Generator Owner/Operator	
Generating Unit	
Month Reporting Data	
Contact Name	
Contact Phone	

Dev. February, 2001
By SPP GWG

Southwest Power Pool

Generator Owner/Operator Voltage Schedule Requirements

Instructions: Control Area Operator shall specify voltage schedule to be maintained by each Generator Owner/Operator's units at a specified bus.

Generator Owner/Operator	
Control Area	
Voltage Schedule Date	
Contact Name (Control Area)	
Contact Phone (Control Area)	

[illegible]

Dev. February, 2001
By SPP GWG

Generator Unit Voltage Schedule Status Report

Generator Owner/Operator	
Generating Unit	
Month Reporting Data	
Contact Name	
Contact Phone	

Dev. February, 2001
By SPP GWG

Southwest Power Pool

Control Area's List of Exempt Generators

Instructions: Control Area Operator shall list units in their area of responsibility that are exempt from following prescribed Control Area Voltage Schedules.

Control Area Operator	
Last Updated	
Contact Name	
Contact Phone	

[illegible]

Dev. February, 2001
By SPP GWG

Generator Unit Transformer Tap Setting Report

Generator Owner/Operator	
Generating Station	
Last Updated	
Contact Name	
Contact Phone	

[illegible]

101

Southwest Power Pool

Transformer Tap Setting Change Request

Instructions: Control Area Operator shall complete information necessary to let affected Generator Owner/Operator know of needed Generator Unit Transformer settings changes.

Generator Owner/Operator	
Generating Station	
Control Area	
Date	
Contact Name (Control Area)	
Contact Phone (Control Area)	

[illegible]

Dev. February, 2001
By SPP GWG

Southwest Power Pool

Generator Units Exempt from Tap Setting Reporting Procedures

Instructions: Generator Owner/Operators shall fill out information for all units under their control that are exempt from Transformer Tap Setting Procedures. The criteria that is met must also be documented

Generator Owner/Operator	
Control Area	
Date	
Contact Name (Control Area)	
Contact Phone (Control Area)	

[illegible]

Dev. February, 2001
By SPP GWG

Southwest Power Pool

Voltage Regulator Control Setting Status Report

Instructions: Generator Owner/Operators shall fill out information for all units under their control.

Generator Owner/Operator	
Generating Unit	
Date	
Contact Name	
Contact Phone	

Generating Control	Control Setting	Generator Short Term Capability	Protective Relay Setting
Overexcitation Limiter			
Underexcitation Limiter			
Volts/Hertz Limiter			

Dev. February, 2001
By SPP GWG

Southwest Power Pool

Generator Governor Characteristic Reporting

Instructions: Generator Owner/Operators shall fill out information for all units under their control.

Generator Owner/Operator	
Generating Unit	
Last Updated	
Contact Name	
Contact Phone	

Governor Control	Control Setting
Speed Regulation	

Dev. February, 2001
By SPP GWG

Southwest Power Pool

Non-Functioning Governor Controls

Instructions: Generator Owner/Operators shall fill out information for all units under their control that has a non-functioning or a blocked speed/load governor control.

Generator Owner/Operator	
Last Updated	
Contact Name	
Contact Phone	

Generating Station	Unit	Governor Status (Blocked/Non-functioning)

Dev. February, 2001
By SPP GWG

12. SPP MISOPERATION REPORT

Reporting Utility: _____ Date: _____

_____ Contact Person: _____

Name	
Phone	
Fax	
Email	
Address	

Equipment Category and Operating Voltage:

DME (Yes/No)	TPS (Yes/No)	GCP (Yes/No)
SPS (Yes/No)	UFLS (Yes/No)	UVLS (Yes/No)
ARL (Yes/No)		Voltage (kV)

Time, Equipment Location & Description:

Incident Date:	Incident Time	Time Zone:
Relay MFG:	Relay Type:	Relay Flags:
Substation Name/Number:	Line/Bus/Auto/Unit Name/No:	Circuit Breaker Numbers:

Description of Misoperation/Failure:

Investigation Results

Corrective Action

Target Date to Complete Recommendations

--

Recommendations

Person verifying that all Corrective Actions and Recommendations have been completed

Name:	Title:	Date:
-------	--------	-------

Note: Equipment categories include DME – Disturbance Monitoring Equip, TPS – Transmission Protection Systems, GCP – Generator Control & Protection, SPS – Special Protection Systems, UFLS – Under Frequency Load Shedding, UVLS – Under Voltage Load Shedding

13. RESPONSE FACTOR THRESHOLDS

Factor	Short-term	Long-term *	Date Approved	Comments
PTDF	3%	3%	4/27/2004	
OTDF	3%	3%	4/27/2004	
PTVF	0.02 p.u.	0.02 p.u.	12/12/2001	
TRM		0%	12/12/2001	See SPP website for short-term TRM
* Long-term factors apply to all elements above 60kV.				

14. INTERCONNECTION REVIEW PROCESS DETAILS

This section shall be subject to review and approval of the Transmission Working Group (TWG). Given the limited amount of time during a regular TWG meeting, and given the need for timely responses, the majority of the interconnection review process will take place outside of a regular meeting. For AC interconnections, a Transmission Assessment Report will be prepared by the parties to the proposed interconnection and presented to the TWG for review. Once received, the TWG will review the information provided in the assessment of impacts on the interconnected system. For HVDC interconnections, the process outlined in the SPP HVDC Planning Manual⁵ will be followed by the parties to the proposed interconnection and presented to the TWG for review in the stages as outlined in Section 14.2.2. Once received, the TWG will review the results from the appropriate study stage (e.g. feasibility, planning or design), and endorse moving to subsequent study stage, performing additional studies, as requested, and/or finalizing review of impacts. This section details the technical requirements for both AC and DC interconnections which shall be the minimum necessary.

14.1 COORDINATION

- 1) The party proposing the interconnection shall appoint a person who will serve as the primary contact with the TWG staff secretary.
- 2) The primary contact shall ensure that all affected parties are identified. This includes but is not limited to notification through the TWG Exploder and requesting any affected parties identify themselves, shall provide a proposed timeline for the studies, shall ensure that all affected parties are notified of and kept informed of progress, including scheduling meetings, as needed, with affected parties, and shall provide the opportunity to review all study results prior to submission to the TWG.
- 3) The primary contact shall coordinate any joint studies that may be necessary and shall report the results to all affected parties.
- 4) All affected parties shall cooperate in joint planning efforts.
- 5) All affected parties will work together to develop an estimated timeline for the completion of the study.
- 6) SPP shall coordinate activities that affect other regions pursuant to an applicable seams agreement.

14.2 PRIOR TO THE REVIEW

14.2.1 AC INTERCONNECTIONS

Affected parties shall jointly develop and evaluate both the proposed interconnection and any mitigation plans. The primary contact shall submit a request for review of the interconnection

⁵ SPP HVDC Planning Manual: <https://spp.org/Documents/74919/SPP%20HVDC%20Planning%20Manual.pdf>

request to the TWG through the group's staff secretary. The request for review shall include the following:

- 1) A list of all affected parties and the contact person at each affected party. The rationale for determining affected parties shall be included.
- 2) A summary of the results of planning studies. Each affected party shall provide a copy of its own planning criteria as documentation of the need of mitigations that exceed regional requirements.
- 3) A detailed description of the project including: in-service date; design information; ratings of the interconnection, a geographic map of the interconnection area, and electrical one-line diagrams of the facilities being connected.
- 4) A summary of the results of power flow, short-circuit, and dynamic analyses in accordance with NERC Reliability Standards, SPP Criteria, other regional requirements, and affected party planning criteria.
- 5) Appropriate program files and program automation files to allow SPP staff to reproduce the studies performed.
- 6) Details of any required mitigation plans including identification of the parties responsible for mitigation. The detailed description of mitigation plans shall include such information as detailed in Item 3 above.
- 7) Any comments of the affected parties.

14.2.2 **HVDC INTERCONNECTIONS**

The primary contact along with the affected parties shall jointly develop and evaluate both the proposed interconnection and any mitigation plans as outlined below. The primary contact shall submit a SPP HVDC Interconnection Request,⁶ including but not limited to Site Control evidence and deposits documents⁷ to begin the study process for the interconnection to the TWG through the group's staff secretary. The initial request, as well as the overall study process, will be outlined and agreed to by SPP and the primary contact in the SPP HVDC (HVDC) Interconnection Study Agreement⁸, which shall include the following:

- 1) A list of all affected parties and the contact person at each affected party. The rationale for determining affected parties shall be included.
- 2) For the initial TWG review prior to beginning the study process, the primary contact will provide:
 - (a) An overview of the project and any known assumptions. Assumptions will include the type of configuration, proposed points of interconnection (POIs), rated capacity and voltage level, length of lines, if applicable, and technologies being utilized.
 - (b) A detailed description of the project, including in-service date, design information, ratings of the interconnection, a geographic map of the interconnection area, and any electrical one-line diagrams of the facilities being connected, as available.

⁶ SPP HVDC Interconnection Request [Link]

⁷ SPP HVDC Planning Manual, Section 1

⁸ SPP HVDC Interconnection Study Agreement [Link]

- 3) Each affected party shall provide a copy of its own planning criteria as documentation of the need of mitigations that exceed regional requirements, as applicable, during the study process.
- 4) The HVDC interconnection study process includes three study stages: Feasibility, Planning and Design.⁹
- 5) Each study stage requires a scope of work (scope) be prepared with the primary contact and reviewed with the affected parties and SPP.¹⁰
 - (a) The primary party will schedule ad hoc meetings with affected parties and SPP, as needed, to review the scope prior to the TWG meeting requesting endorsement.
 - (b) The primary contact will submit a request for review of the study stage scope to the TWG through the group's staff secretary.
 - (c) The TWG will review and approve each final study stage scope outlining the models and analyses required for that study stage, as outlined in Section 14.3.2.
 - (d) Additional project-specific studies may be requested in accordance with Section 14.3.¹¹
- 6) The primary contact shall submit a request for review of each of the study stage results to the TWG through the group's staff secretary. In addition, the primary party will:
 - (a) Provide a summary of the results of the analyses performed for each study stage.
 - (b) Schedule ad hoc meetings with affected parties and SPP, as needed, to review the study stage results prior to the TWG meetings requesting endorsement.
 - (c) Submit a request for review of the study stage results to the TWG through the group's staff secretary.
 - (d) Present each final study stage results to the TWG for review and endorsement, as outlined in Section 14.3.2.
 - (e) Provide appropriate program files and program automation files to allow SPP staff to reproduce the studies performed, as needed and/or requested.
 - (f) Describe any required mitigation plans including identification of the parties responsible for mitigation. The detailed description of mitigation plans shall include such information as detailed in Section 14.2.2, item 2(b) above.
 - (g) Any comments of the affected parties.

14.3 TECHNICAL STUDY REQUIREMENTS

This section outlines the technical study requirements for both AC and HVDC interconnections. The TWG may request additional studies at its discretion. The rationale for requesting additional studies shall be provided to all affected parties.

14.3.1 AC INTERCONNECTIONS

Impact analysis shall be performed to assess the system with and without the interconnection to determine if the interconnection causes any adverse effects to the system.

⁹ SPP HVDC Planning Manual, Section 3.

¹⁰ SPP HVDC Planning Manual, Section 3.

¹¹ SPP HVDC Planning Manual, Section 9.1.

The following are minimum requirements for power flow analysis:

- 1) A contingency assessment consistent with Integrated Transmission Planning Manual in the entire first-tier area of the combined areas which the proposed interconnection connects. This area may be expanded or reduced as mutually agreed upon. The rationale for expansion or reduction of the study area shall be documented and agreed upon by all affected areas.
- 2) A review of impacts shall utilize all applicable Scenario Cases developed by the SPP extending to the planning horizon year.
- 3) If, at any time, impacts are identified affecting a nuclear power plant, it shall be included separately as an affected party.
- 4) Affected parties shall report adverse impacts and required mitigations.

The TWG may request additional studies at its discretion. The rationale for requesting additional studies shall be provided to all affected parties. If the proposed interconnection was previously evaluated by SPP and included in the most recent SPP Transmission Expansion Plan (STEP), the power flow analysis described here does not need to be repeated.

The following are minimum requirements for short-circuit analysis:

- 1) Analysis shall be consistent with the Integrated Transmission Planning Manual. The model(s) assessed shall be determined by the affected parties. The rationale for the model(s) assessed shall be included.
- 2) Assessment shall consist of three-phase and phase-to-ground faults applied at the buses of the proposed interconnection plus all first-tier buses to the interconnection.
- 3) Additional buses may be studied as mutually agreed upon and documentation for including such additional buses shall be included.
- 4) Assessments shall document the before and after fault currents on all monitored buses.
- 5) Affected parties shall report adverse impacts and the required mitigation.

The following are minimum requirements for transient stability analysis:

- 1) Analysis shall be performed using a contingency assessment. The model(s) and contingencies assessed shall be determined by the affected parties. Consideration shall be applied for evaluating impactful contingent events from neighboring SPP or non-SPP systems. The rationale for the model(s) and contingencies assessed shall be included. If no transient stability analysis is performed the rationale for not performing such studies shall be provided.
- 2) The assessments performed shall be determined by the affected parties. The rationale for the assessments shall be included.
- 3) Assessments shall document the before and after post disturbance performance on all monitored buses consistent with Section 5.4 of this Criteria.
- 4) Affected parties shall report adverse impacts and the required mitigation.

If the interconnection is to be made at 345 kV or higher voltage, affected parties shall determine if additional study work is needed to assess reactive power impacts and management. If the affected parties determine additional study work is required, results of these studies shall be provided for review. Additional study work may include but is not limited to steady state, transient (including electromagnetics transient studies), surge impedance loading for long lines, insulation coordination, or switching studies clearly indicating any required levels of shunt compensation. If no additional study work is performed for 345 kV or higher interconnections, the rationale for not performing such studies shall be provided.

14.3.2 **HVDC INTERCONNECTIONS**

The technical study process HVDC interconnections include the feasibility, planning, and design stages, as well as project-specific studies, if applicable. These studies shall be done by HVDC Owner and shall be reviewed by SPP based on the SPP HVDC Planning Manual. Any deviations from the study stage processes in the SPP HVDC Planning Manual will be communicated to SPP, the affected parties, and approved by TWG.¹²

14.4 **DISPUTE RESOLUTION**

All disputes between SPP members shall be resolved using the procedures of Section 3.13 of the SPP Bylaws.

14.5 **REVIEW AND BALLOT BY THE TRANSMISSION WORKING GROUP**

- 1) The TWG Secretary shall review the request for interconnection for completeness.
- 2) Any deficiencies shall be reported to the primary contact.
- 3) Once a complete request is received, it shall be forwarded to the TWG for a 14 day review and comment period.
- 4) The primary contact shall be responsible for coordinating any response necessary to comments and questions raised by the TWG.
- 5) The TWG shall schedule a ballot to accept or to reject the interconnection request.
- 6) Any action taken by the TWG will be included in the minutes.

14.5.1 **TRANSMISSION INTERCONNECTION REVIEW DATA CHECKLIST**

- 1) Primary contact and all affected parties' contact information.
- 2) Overview of the proposed interconnection and its need.
- 3) Estimated or proposed in-service date.
- 4) List of all studies run by season.
 - (a) AC Interconnection
 - (i) Power flow studies minimum requirements met.
 - (ii) Short-circuit studies minimum requirements met.
 - (iii) Transient studies minimum requirements met.
 - (b) HVDC Interconnections
 - (i) Minimum study requirements met based on SPP HVDC Planning

¹² SPP HVDC Planning Manual, Section 3.

Manual.¹³

- 5) Affected parties planning criteria, if applicable.
- 6) A detailed description of the proposed interconnection.
 - (a) In- service date
 - (b) Design information
 - (c) Ratings of the interconnection
 - (d) A geographic map of the interconnection area
 - (e) Electrical one-line diagrams of the facilities being connected.
- 7) Appropriate program files and program automation files to allow SPP staff to reproduce the studies performed.
- 8) Details of any required mitigation plans including identification of the affected parties responsible for mitigation.
 - (a) In-service date
 - (b) Design information
 - (c) Ratings of the facilities
 - (d) A geographic map of the facility area
 - (e) Electrical one-line diagrams of the facilities being connected.
- 9) Comments of affected parties covering agreement or points of disagreement of the proposed interconnection, if any.

The TWG shall review and modify this section as needed but not less frequently than once every three years.

¹³ SPP HVDC Planning Manual.

15. APPENDIX 1 - SPP RAS THRESHOLDS

15.1 PURPOSE

A Remedial Action Scheme (RAS) is designed to detect predetermined system conditions and automatically take corrective actions that may include, but are not limited to, adjusting or tripping generation (MW and Mvar), tripping load, or reconfiguring a system(s). This appendix establishes the extents of instantaneous action to remove load or generation by a RAS allowed on the SPP transmission system.

15.2 RAS THRESHOLDS

No SPP RAS shall instantaneously remove load or generation exceeding the thresholds given in Table 15.1.

SPP RAS Type	Load Threshold [MW] ¹⁴	Generation Threshold [MW]
Extended Use RAS	ZPC	600
Temporary RAS	ZPC	600

Table 15.1 – SPP RAS Instantaneous Removal Thresholds

A further non-instantaneous control action to reduce generation or load by the RAS as part of its designed operation may be allowed.

The RAS generation thresholds shall be reviewed annually to ensure they are aligned with SPP's spinning reserve requirements.

¹⁴ If no threshold related to RAS load shed is established in a documented ZPC accepted by SPP, the threshold would default to 200 MW.